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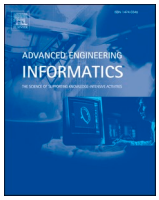

Full length article

Cross-domain adaptive digital twin modelling for dam structures using graph attention networks

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ABSTRACT

Digital twin technology has emerged as a promising paradigm for advancing instrumentation and measurement in dam safety monitoring. However, existing models often face significant cross-domain discrepancies between numerical simulations (source domain) and in-situ measurement data (target domain), which constrain prediction accuracy and generalization under varying operating conditions. To address these challenges, this study proposes a novel cross-domain adaptive digital twin framework that integrates graph attention networks, gating mechanisms, and feature alignment strategies to enhance measurement consistency and predictive reliability. Multi-source numerical simulation domains are systematically constructed to enrich input variability and improve measurement-domain coverage. The proposed framework is comprehensively validated through ablation studies and comparisons with six state-of-the-art baseline models, achieving superior accuracy with mean errors as low as 0.44 mm and 0.99 %, thereby demonstrating excellent measurement precision. Furthermore, validation under extreme water-level and temperature scenarios confirms strong generalization, with the inclusion of extreme-condition samples further reducing errors and mitigating prediction degradation at critical monitoring points. The results show that the proposed approach provides a physically consistent, generalizable, and robust digital twin measurement framework for dam structures, enabling reliable real-time deformation monitoring and safety evaluation. These findings establish a foundation for intelligent, large-scale, and multi-condition applications of digital twins in civil infrastructure monitoring and measurement science.

1. Introduction

With the continuous advancement of digital technologies driving the transformation and upgrading of hydraulic engineering, the construction of high dams is facing unprecedented technical challenges and demands [1,2]. Along with the in-depth implementation of China's Western Development Strategy, a series of ultra-high dams have been successively constructed and put into operation, such as the Jinping-I Arch Dam (305 m), the Lianghekou Dam (292 m), the Baihetan Arch Dam (279 m), and the Shuangjiangkou Dam (305 m) [3,4]. These projects are characterized by massive dam volumes, enormous reservoir capacities, and high stress levels, which impose stringent requirements on real-time monitoring and safety assessment [5–7]. However, as dam structures become increasingly complex and large in scale, the limitations of traditional monitoring and management approaches in terms of data acquisition, information integration, and state prediction are becoming more prominent. Therefore, the adoption of advanced digital twin technology to achieve comprehensive real-time monitoring and in-depth data analysis is expected to provide strong support for accurate perception of structural states, enhanced safety assurance, and optimized operational decision-making [8,9].

Structural Health Monitoring (SHM) methods have been widely applied in the safety evaluation of dams and can be generally classified into data-driven and mechanism-driven models [10]. Data-driven approaches, including Hydrostatic-Seasonal-Time (HST) models and machine learning techniques, focus on statistical correlations, while mechanism-driven models rely on finite element analysis and reliability theory to assess structural risk [11]. Recent advances in deep learning have enhanced the modelling capacity of traditional HST methods [12], with techniques such as clustering, support vector

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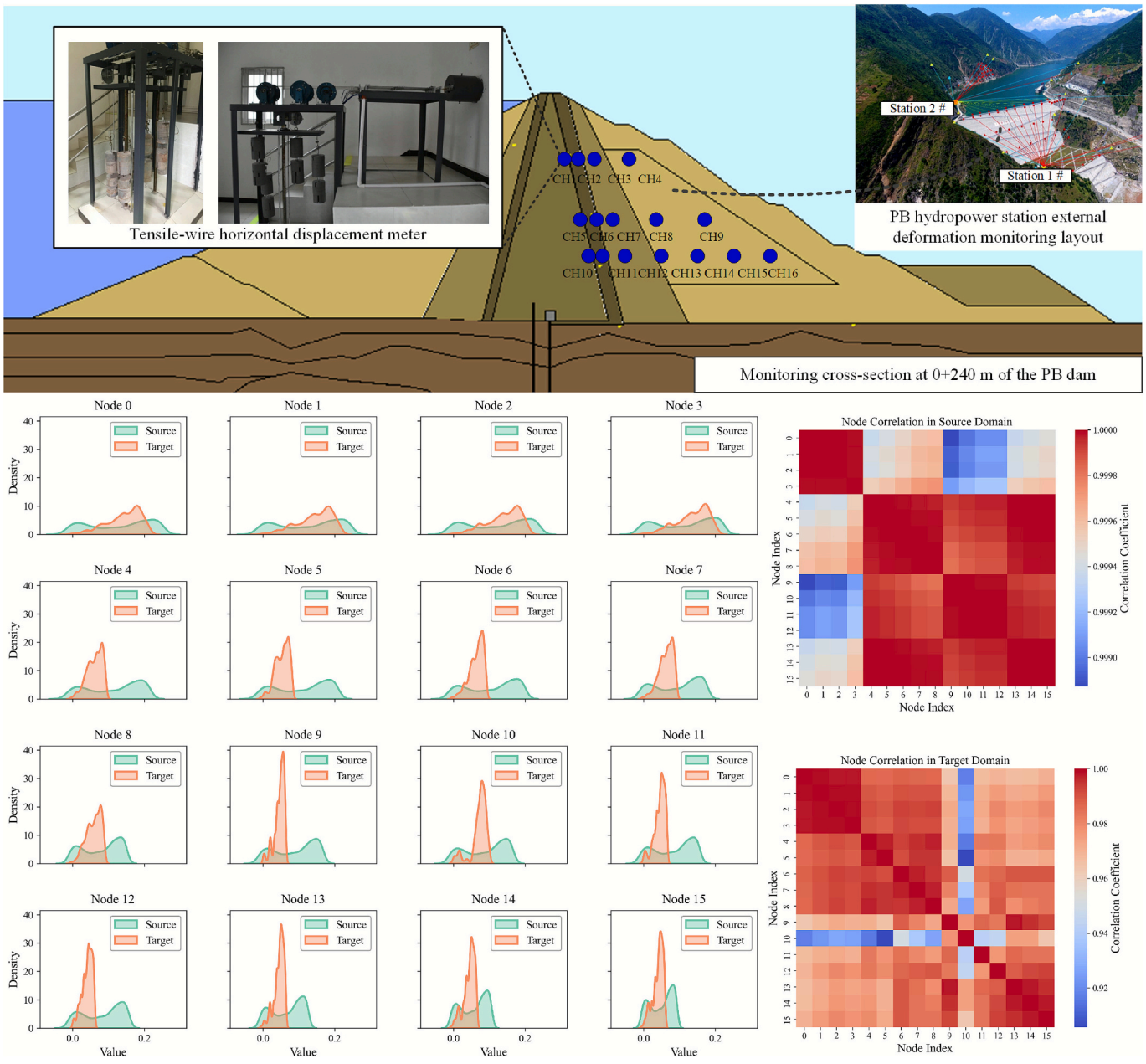

Fig. 1. Utilize the kernel density method and spatial correlation analysis to visualize the cross-domain differences between the numerical simulation source domain and on-site measurement.

machines, and extreme learning machines applied to spatiotemporal segmentation and displacement prediction [13,14]. However, shallow models struggle to capture complex nonlinear and temporal dependencies. Deep networks such as Long Short-Term Memory Network (LSTM) [15] and Graph Convolutional Networks (GCNs) [16] can better represent temporal and spatial correlations but remain constrained by data availability and interpretability. Consequently, existing data-driven approaches are limited to localized monitoring points and cannot fully reflect the global behavior of large-scale dams, underscoring the need for an integrated digital twin framework that fuses simulation and monitoring data for comprehensive dam safety assessment.

In the context of Industry 4.0, digital twin technology provides a novel solution for global real-time monitoring of dams [17,18]. By integrating Building Information Modelling (BIM) and SHM technologies, digital twins enable the construction of high-fidelity virtual replicas of engineering structures and facilitate the perception of structural states and the prediction of potential damage [19–21]. Through the combination of physical models, sensor data, and artificial intelligence techniques, digital twins realize real-time synchronization of physical system states and trend prediction [22,23]. The core architecture consists of physical entities, digital models, data transmission, real-time interaction, and intelligent analysis [24]. The introduction of deep learning and reinforcement learning endows the system with adaptive learning capabilities and enables intelligent decision-making and optimization [25]. A major advantage of digital twins lies in their ability to perform testing and optimization within a low-cost and low-risk virtual environment, thereby reducing uncertainties during the operation and maintenance of actual engineering projects [26,27].

In recent years, the concept of the digital twin has evolved rapidly across various engineering disciplines, highlighting the integration of physical modelling, numerical simulation, and data-driven learning. As noted by de Wilde, the boundaries between building performance simulation, machine learning, and digital twins are conceptually

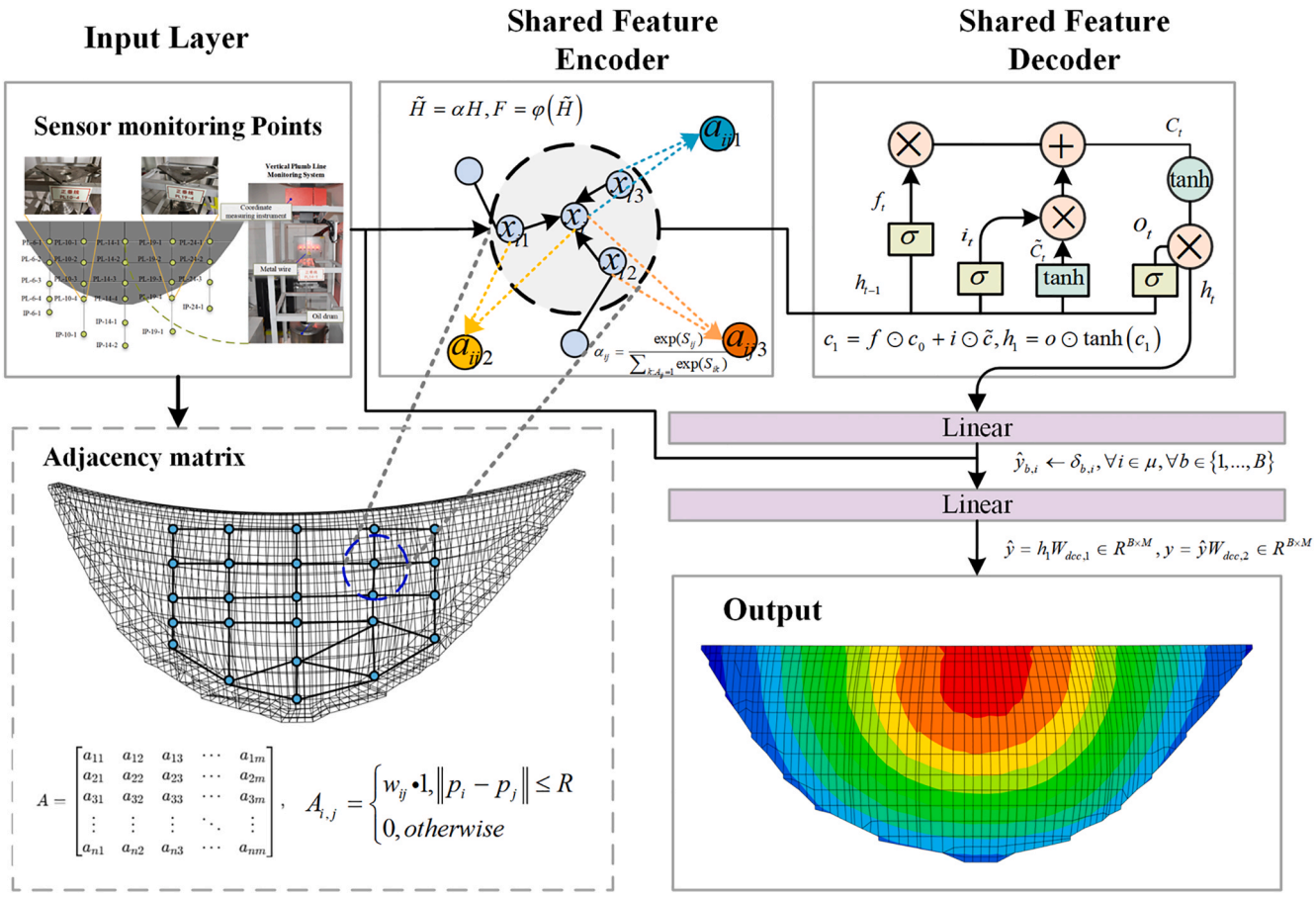

Fig. 2. Proposed method network architecture.

overlapping rather than distinct [28]. A digital twin is characterized by four essential components: modelling and simulation, data fusion, interaction and collaboration, and service, reflecting the continuous information exchange between physical and virtual entities. In civil and structural engineering, digital twin frameworks have been established in railways [29], bridges [30], tunnels [31,32], and mechanical systems [23] through the combination of physics-based finite element simulation and machine learning. This hybrid paradigm enhances predictive accuracy and interpretability.

At present, research on digital twin models for dam structures is still at an early exploratory stage [33]. Existing studies are mainly based on Geographic Information Systems (GIS) and BIM to construct geospatial digital twin models of dams and river systems [34]. Although such methods can achieve high-precision digitalization of engineering structures, their practical application remains limited due to the lack of real-time dynamic responses to environmental changes. Other scholars have applied vision-based stereo camera systems for real-time deformation monitoring of gates. While such approaches provide significant value for external safety assessment [35,36], they fail to capture changes in the internal structural state and therefore offer limited support for comprehensive structural health monitoring [37].

Some studies have constructed digital twin models based on numerical simulation within the Finite Element Method (FEM) [38,39]. However, such approaches often require extensive calibration of material parameters and structural simplifications [40,41], and involve computationally demanding iterative equation solving [42], which constrains both efficiency and accuracy. To overcome these challenges, recent studies have proposed FEM-deep learning-coupled digital twin frameworks for analyzing dam deformation, seepage, and stress evolution, demonstrating the feasibility of physics-informed digital twins for dam safety monitoring [43,44]. These models combine data-driven and physics-based approaches. Nevertheless, their accuracy is still restricted by domain discrepancies between the numerical simulation source domain and the monitoring data target domain, as illustrated in Fig. 1 [45]. Furthermore, the extrapolation and generalization capabilities of data-driven models remain to be validated [44]. Therefore, digital twin methodologies in the field of dam structural safety monitoring still face critical technical bottlenecks, and their construction methods require further investigation.

At present, cross-domain adaptive modelling from numerical simulation source domains to monitoring data target domains in dam digital twins remains largely unexplored. Although studies in related subfields, such as dam image recognition, have attempted to employ transfer learning [46–48] and *meta*-learning [49] to achieve domain adaptation and generalization to mitigate data scarcity, existing work has primarily focused on pre-training and fine-tuning frameworks for cross-site crack detection [50,51] or condition monitoring [52,53]. Due to the absence of quantitative characterization of cross-domain discrepancies and high-dimensional feature alignment, the applicability of these methods to cross-domain modelling in digital twins is still limited [54].

In summary, current digital twin models of dam structures are primarily constructed based on numerical simulation and deep learning. The accuracy of these models is affected by cross-domain discrepancies between simulation-based source data and monitoring-based target data. At the same time, the extrapolation and generalization capacity of deep learning approaches remains uncertain. To address these issues, the present study proposes a cross-domain adaptive method to mitigate discrepancies between source and target domains, thereby improving the accuracy of digital twin models established through numerical simulation and deep learning. In addition, the proposed method is

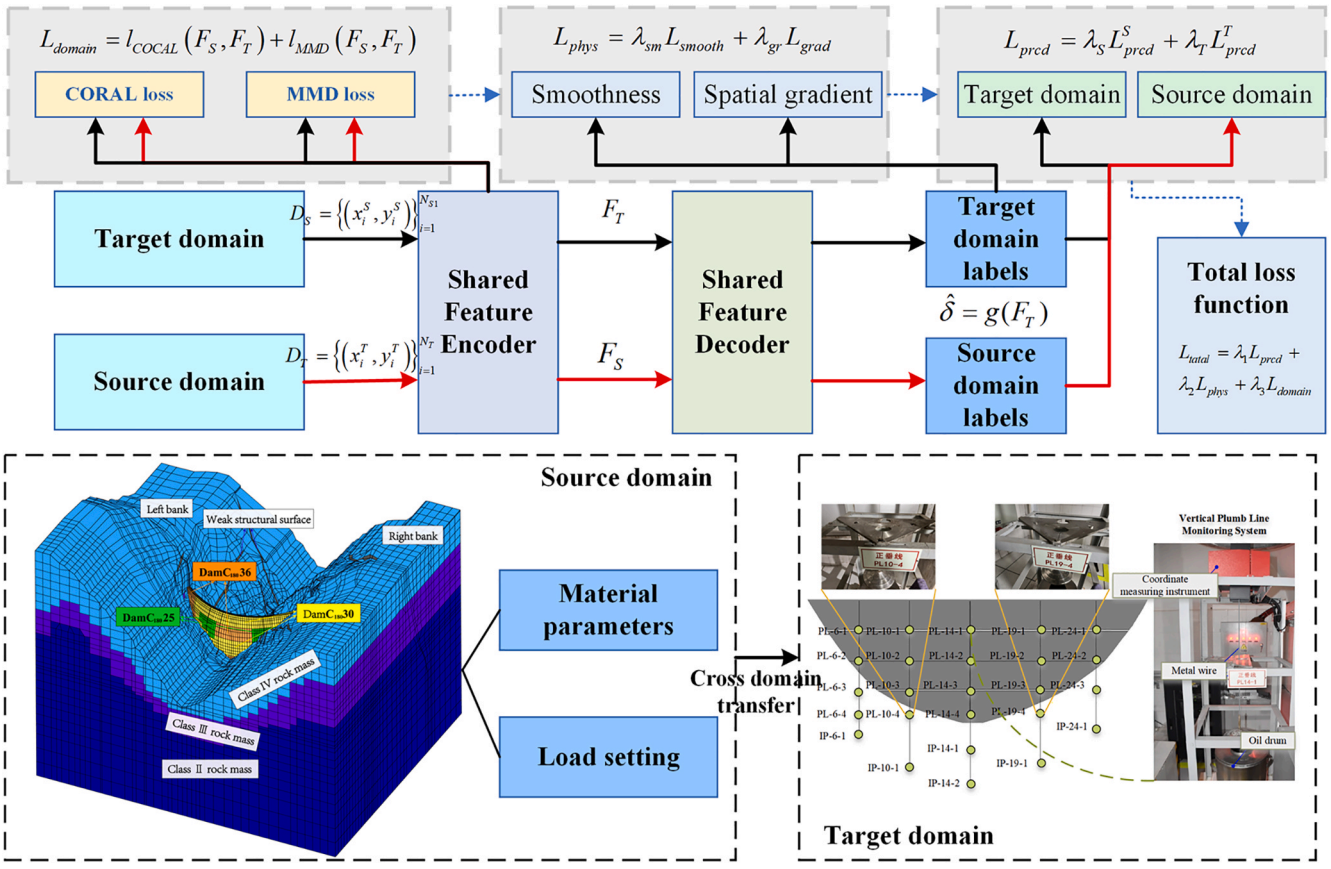

Fig. 3. The proposed cross domain adaptive modelling architecture.

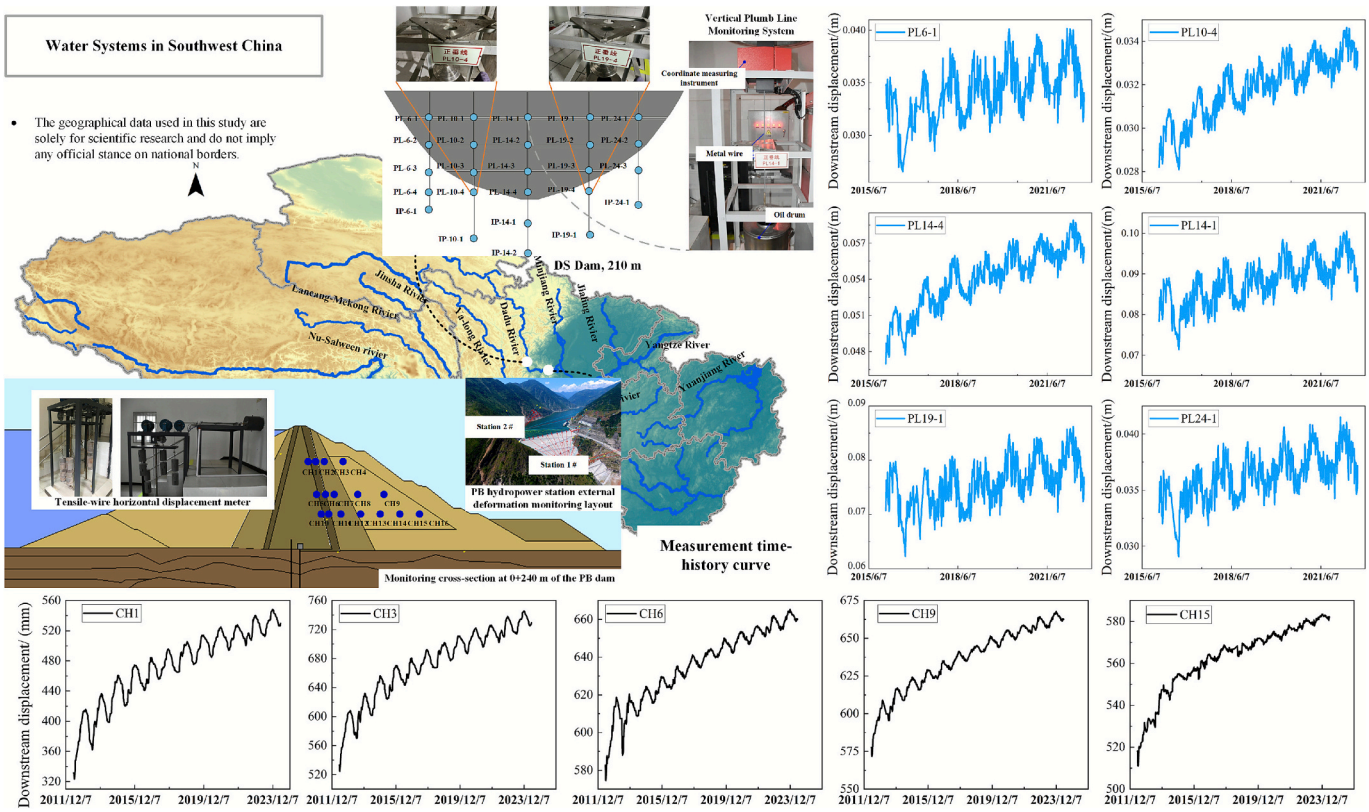

Fig. 4. Layout of deformation monitoring and longitudinal displacement process lines for DS dam and PB dam.

Table 1
Datasets specifics table.

Dams	Datasets	Period	Length	Input dimension	Output points
DS	Source Domain I	2016–2022	2437	18	4902
	Source Domain II	2016–2022	2437	18	4902
	Source Domain III	2016–2022	2437	18	4902
	Source Domain IV	2016–2022	2437	18	4902
	Source Domain V	2016–2022	2437	18	4902
	Target Domain	2016–2022	2437	18	4902
PB	Source Domain	2012–2024	4380	16	3238
	Target Domain	2012–2024	4380	16	3238

validated under extreme water level and temperature conditions to evaluate its generalization capability. The main contributions of this study are as follows:

- A refined theoretical framework for digital twin modelling of dam structures based on deep transfer learning and the finite element method is established. A new source-domain construction method and cross-domain adaptive strategy are proposed to enhance modelling accuracy and generalization under extreme operating conditions.
- A novel graph attention-gated network architecture is designed to improve the interpretability and predictive reliability of the digital twin model.
- The proposed framework is applied to the high arch dam and high earth-rock dam, and its effectiveness is validated through ablation experiments and comparisons with multiple baseline models. Furthermore, the model's generalization performance under extreme water-level and temperature conditions and the influence of neural network architectures on prediction accuracy are systematically analyzed.

2. Proposed method

This section proposes a digital twin modelling approach for dam structures. First, a digital twin model network is established based on Graph Attention Networks (GATs) and gating mechanism, and the adjacency matrix is constructed according to the arrangement of measurement points. Subsequently, numerical simulations are generated as the source domain based on material parameters and various operating conditions. Cross-domain learning is then achieved through Correlation Alignment (CORAL) and Maximum Mean Discrepancy (MMD) feature alignment between the numerical simulation source domain and the

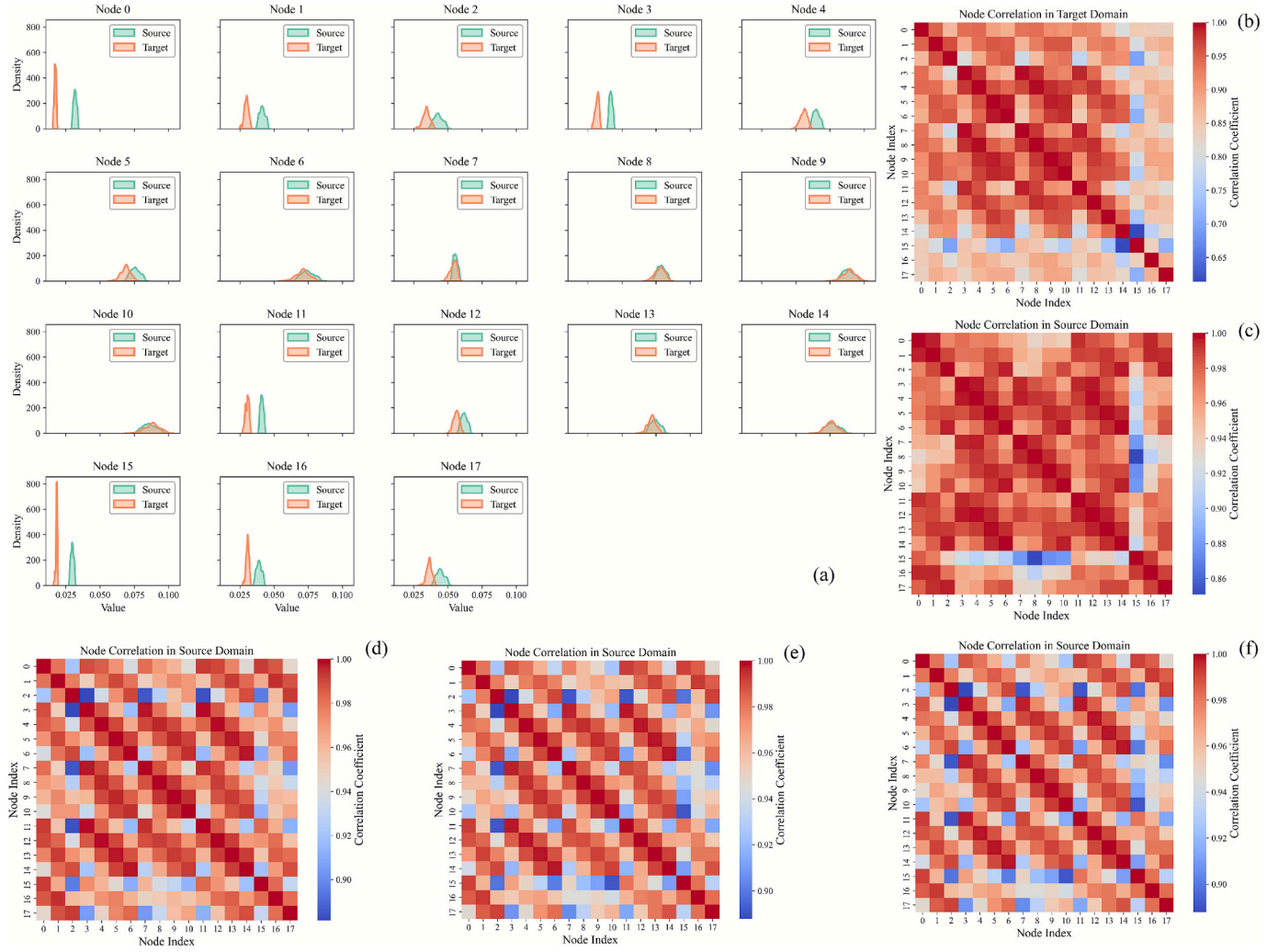

Fig. 5. Visualization of adaptability between DS dam source domain and target domain. (a) Kernel density estimation (b) Target domain (c) Source Domain I (d) Source Domain II (e) Source Domain III (f) Source Domain IV.

Table 2
The main parameter settings of the models in the paper.

Model	Main Parameter Setting
Proposed model	hidden layer dimension = 32
Ablation Model I	hidden layer dimension = 32
Ablation Model II	hidden layer dimension = 32
Ablation Model III	hidden layer dimension = 32
Ablation Model III	hidden layer dimension = 192
LSTM	Hidden layer dimension = 192
GRU	Hidden layer dimension = 192
Transformer	hidden layer dimension = 192, encoder layer = 2, decoder layer = 2, attention head = 4
GCN	Hidden layer dimension = 192
TGCN	Hidden layer dimension = 192
GAT	Hidden layer dimension = 192, attention head = 4

monitoring data target domain, enabling the digital twin representation of the high arch dam structure.

2.1. Neural network architecture

To efficiently model the spatial correlations among dam monitoring points and to selectively preserve or suppress key information, this study proposes a two-stage graph attention and gating digital twin neural network architecture, as shown in Fig. 2. The input comprises single-instance (or simultaneous) multi-point observations, which are first processed by a graph attention layer [55] for spatial encoding. The encoded results are subsequently processed by a gating unit, implemented with LSTM [56] in a single-step manner, to perform a one-time memory and forgetting operation. This process produces the reconstructed full-field features, as illustrated in the figure.

The feature encoder $\phi(\bullet)$ is constructed based on a graph attention layer. Suppose there are N monitoring points, and the input is denoted as $x \in R^{B\times N}$ with a batch size of B. The graph attention layer first performs a linear transformation:

$$H = xW \tag{1}$$

where W represents the learnable weights and is implemented using a bias-free linear layer. The aggregation is performed on the given graph structure $A \in \{0,1\}^{N\times M}$, which is designed to reflect both the force/strain propagation range in the context of material mechanics and the geometric relationships among the monitoring points. An inner-product-based attention scoring mechanism is defined, and masking is applied to the zero connections:

$$S = (HH)A, S_{ij} = \begin{cases} \langle h_i, h_i \rangle, A_{ij} = 1 \\ -\infty \end{cases} \tag{2}$$

where h_i represents the encoded feature of the i-th monitoring point. Subsequently, a softmax normalization is applied over the neighborhood of each i to obtain the attention coefficients [57]:

$$\alpha_{ij} = \frac{\exp(S_{ij})}{\sum_{k:A_{ij}=1}\exp(S_{ik})}, \sum_j \alpha_{ij} = 1 \tag{3}$$

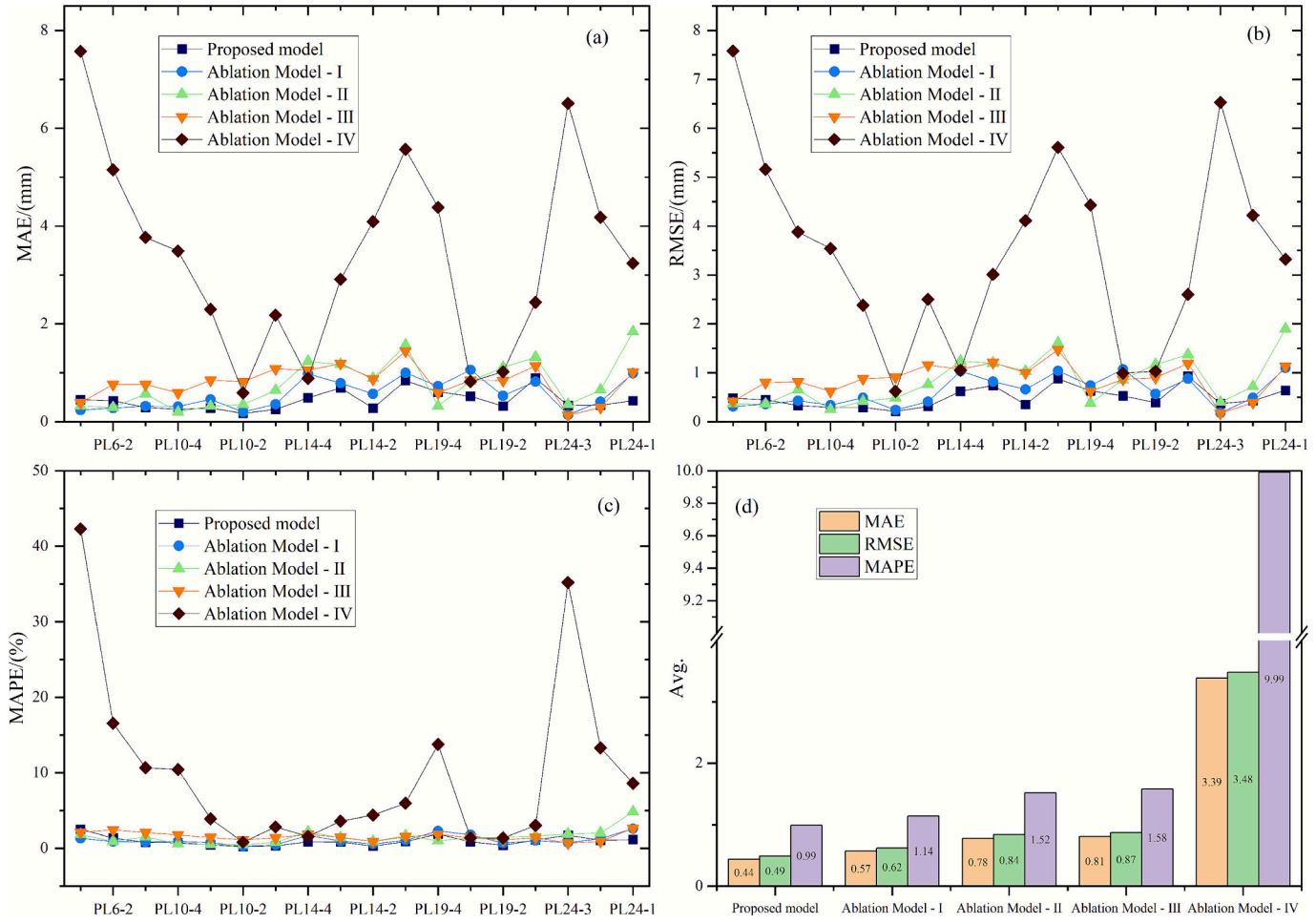

Fig. 6. Compare the MAE, RMSE, and MAPE of the proposed model with the ablation models at all measurement points.

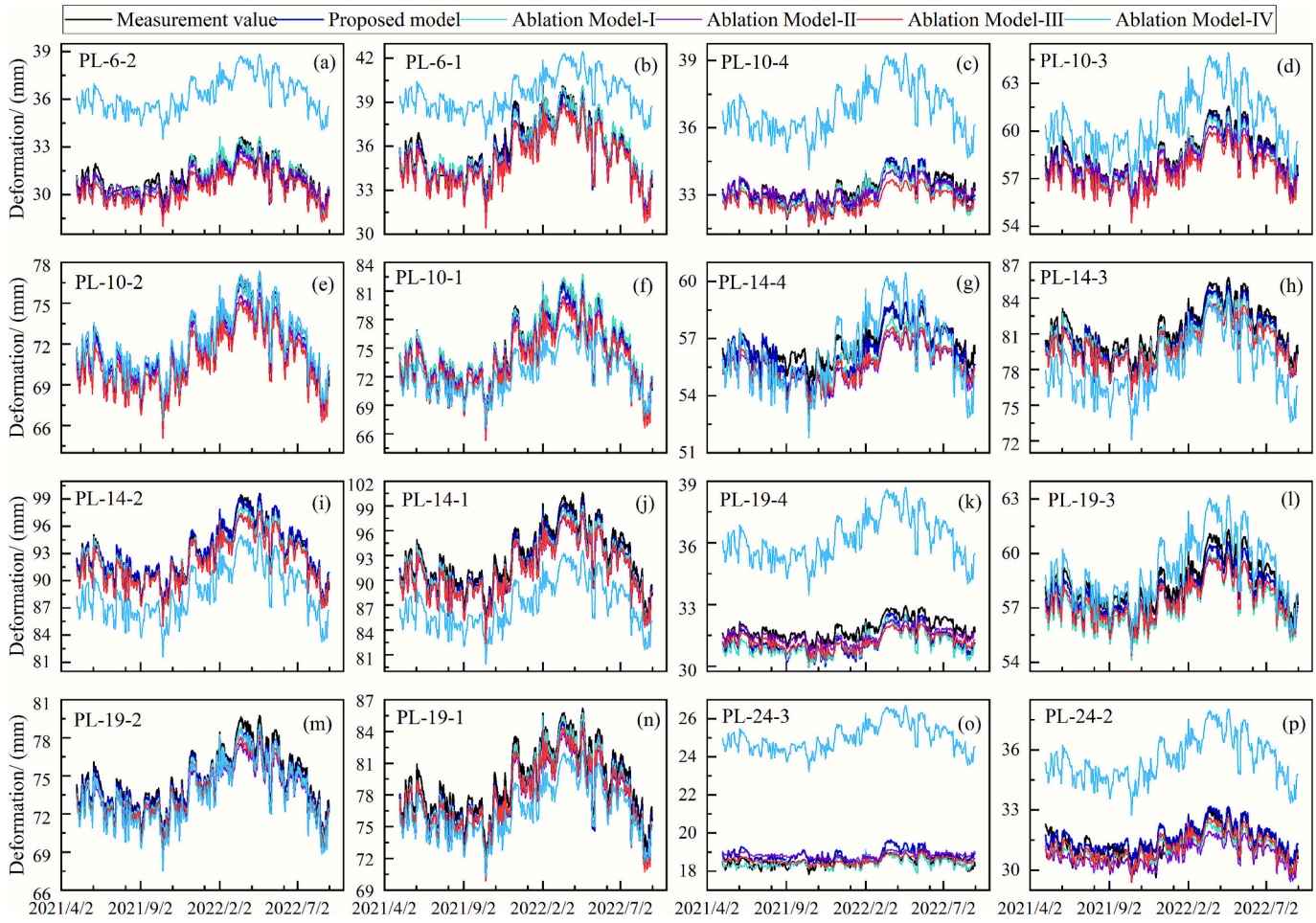

Fig. 7. Comparison of the measurement process between the proposed model and the ablation models on the test set.

The aggregation yields the feature-encoded output F:

$$\widetilde{H} = \alpha H, F = \varphi\left(\widetilde{H}\right) \tag{4}$$

where $\varphi(\bullet)$ denotes the activation function, which is implemented using the Exponential Linear Unit (ELU).

Let $v = \{1, ..., M\}$ denote the set of finite element nodes (including monitoring points), with spatial coordinates represented by $\{p_i \in R^3\}$. According to the spatial attenuation law of stress–strain in material mechanics, an influence radius R is introduced to control edge connections, and a distance-based weighting function is employed to describe the edge weights. The weighted adjacency matrix A is then defined as:

$$A_{i,j} = \begin{cases} w_{ij} \bullet 1, \left\|p_i - p_j\right\| \leqslant R \\ 0, otherwise \end{cases} \tag{5}$$

$$w_{ij} = \exp\left(-\frac{\left\|p_i - p_j\right\|}{\kappa}\right) \tag{6}$$

where w_{ij} denotes the distance decay weight, κ represents the propagation characteristic length associated with the material and structural scales (which can be calibrated using the elastic modulus, structural dimensions, or numerical sensitivity), and R controls both the sparsity of the graph and the physical adjacency.

To perform the feature decoder $g(\bullet)$ on the encoded feature, the feature encoding F is processed through a single-step LSTMCell for gating. The initial hidden state and cell state are set to $h_0 = 0, c_0 = 0$, and the four gates of the LSTMCell are computed as follows [58]:

$$i = \sigma(FW_i + b_i) \tag{7}$$

$$f = \sigma(FW_f + b_f) \tag{8}$$

$$o = \sigma(FW_o + b_o) \tag{9}$$

$$\widetilde{c} = \tanh(FW_c + b_c) \tag{10}$$

The single-step update is expressed as follows:

$$c_1 = f \odot c_0 + i \odot \widetilde{c}, h_1 = o \odot \tanh(c_1) \tag{11}$$

The result is then linearly mapped to an $M-$dimensional monitoring point space:

$$\widehat{y} = h_1 W_{dcc,1} \in R^{B \times M}, y = \widehat{y} W_{dcc,2} \in R^{B \times M} \tag{12}$$

To enhance the fusion of high-dimensional and low-dimensional features, the set of monitoring point indices $\mu \subset \{1, ..., N\}$ is used with an observation coverage strategy. In the intermediate reconstruction $\widehat{y}$, the original observations are used to cover the corresponding positions [44]:

$$\widehat{y}_{b,i} \leftarrow \delta_{b,i}, \forall i \in \mu, \forall b \in \{1, ..., B\} \tag{13}$$

2.2. Source domain construction

In this study, the numerical simulation data obtained from finite element analysis are defined as the source domain $D_S = \left\{\left(x_S^i, y_S^i\right)\right\}_{i=1}^{N_S}$,

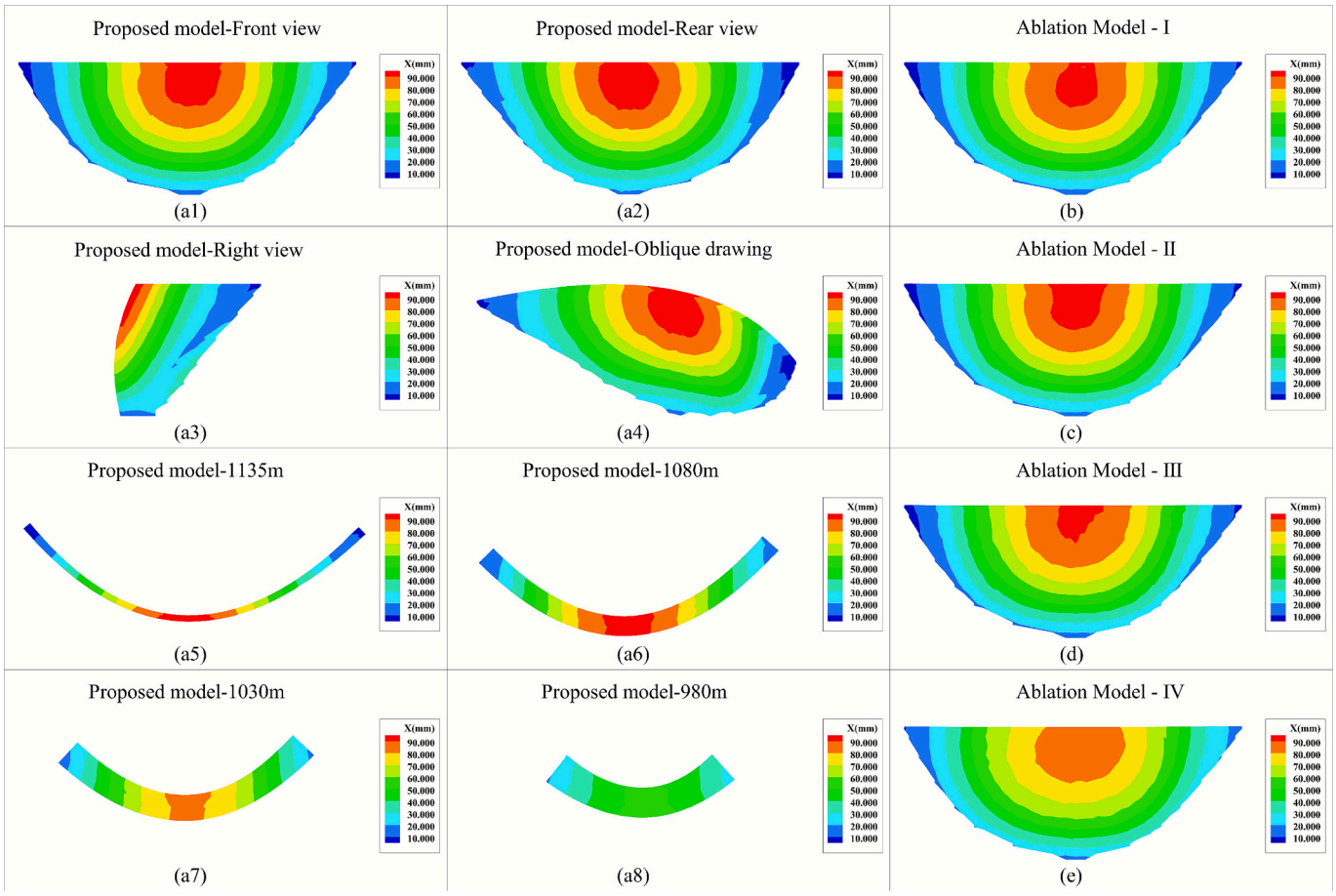

Fig. 8. Comparison of 3D visualization between the proposed model and the ablation models.

while the actual monitoring data are defined as the target domain $D_T = \{(x_T^i, y_T^i)\}_{i=1}^{N_T}$.Due to differences in their data generation mechanisms, the two domains exhibit significant discrepancies in feature distribution, temporal scale, and physical consistency. This domain gap can be expressed as the distribution divergence between the marginal and conditional probabilities:

$$\Delta_{domain} = D(P_S(x), P_T(x)) + D(P_S(y|x), P_T(y|x)) \tag{14}$$

where $D(\cdot,\cdot)$ denotes a statistical distance measure (e.g., Maximum Mean Discrepancy, Kullback–Leibler divergence). The goal of cross-domain adaptive modelling is to minimize Δ_{domain} through feature-level alignment and multi-source enhancement.

The formation mechanism of domain discrepancies mainly originates from two aspects: (1) numerical simulation assumptions, including material homogeneity and idealized boundary conditions, which ensure theoretical precision but weaken representation of real operation states; and (2) monitoring uncertainty, arising from uneven sensor placement, environmental noise, and measurement bias, which distort the actual feature distribution. To improve digital twin fidelity, two strategies are considered.

High-fidelity source-domain reconstruction. The finite element model is refined to approach actual engineering conditions by optimizing geometric parameters G and material parameters M. The optimization problem can be formulated as:

$$\min_{G,M} \| y_S(G,M) - y_T \|_2^2 \tag{15}$$

where $y_S(G,M)$ represents simulated responses, and y_T denotes the measured deformation or stress.

Multi-source domain construction and cross-domain adaptation. To overcome the limited coverage of a single source domain, a multi-source hybrid domain $\{D_S^k\}_{k=1}^K$ is constructed by perturbing key physical parameters (material, load, and boundary conditions). Each sub-domain corresponds to a different scenario:

$$D_S^k = \{x_S^i(\theta_k), y_S^i(\theta_k)\}, \theta_k = \{M_k, B_k, L_k\} \tag{16}$$

where M_k, B_k and L_k denote the material properties, boundary conditions, and loading states for the k-th domain.

2.3. Cross-domain adaptive modelling

To further enhance the modelling performance digital twin model of dam structures, this study proposes a transfer learning framework based on domain adaptation [59–61], as illustrated in Fig. 3. Specifically, the proposed method employs the Shared Feature Encoder (SFE) $\phi(\bullet)$ to extract domain-representative features F_S and F_T, followed by COCAL and MMD to learn domain-invariant representations. The mathematical formulations are as follows [62,63]:

$$l_{COCAL}(F_S, F_T) = \frac{1}{4d^2} \| \frac{1}{N_S - 1}\left(F_S^T F_S - \mu_S^T \mu_S\right) - \frac{1}{N_T - 1}\left(F_T^T F_T - \mu_T^T \mu_T\right) \|_F^2 \tag{17}$$

$$l_{MMD}(F_S, F_T) = \left\| \frac{1}{N_S}\sum_{i=1}^{N_S} \Phi\left(f_S^i\right) - \frac{1}{N_T}\sum_{i=1}^{N_T} \Phi\left(f_T^i\right) \right\|^2 \tag{18}$$

where μ_S, μ_T represents the mean vectors of the corresponding features F_S, F_T, $\| \bullet \|_F$ is the Frobenius norm, d denotes the dimensionality of the

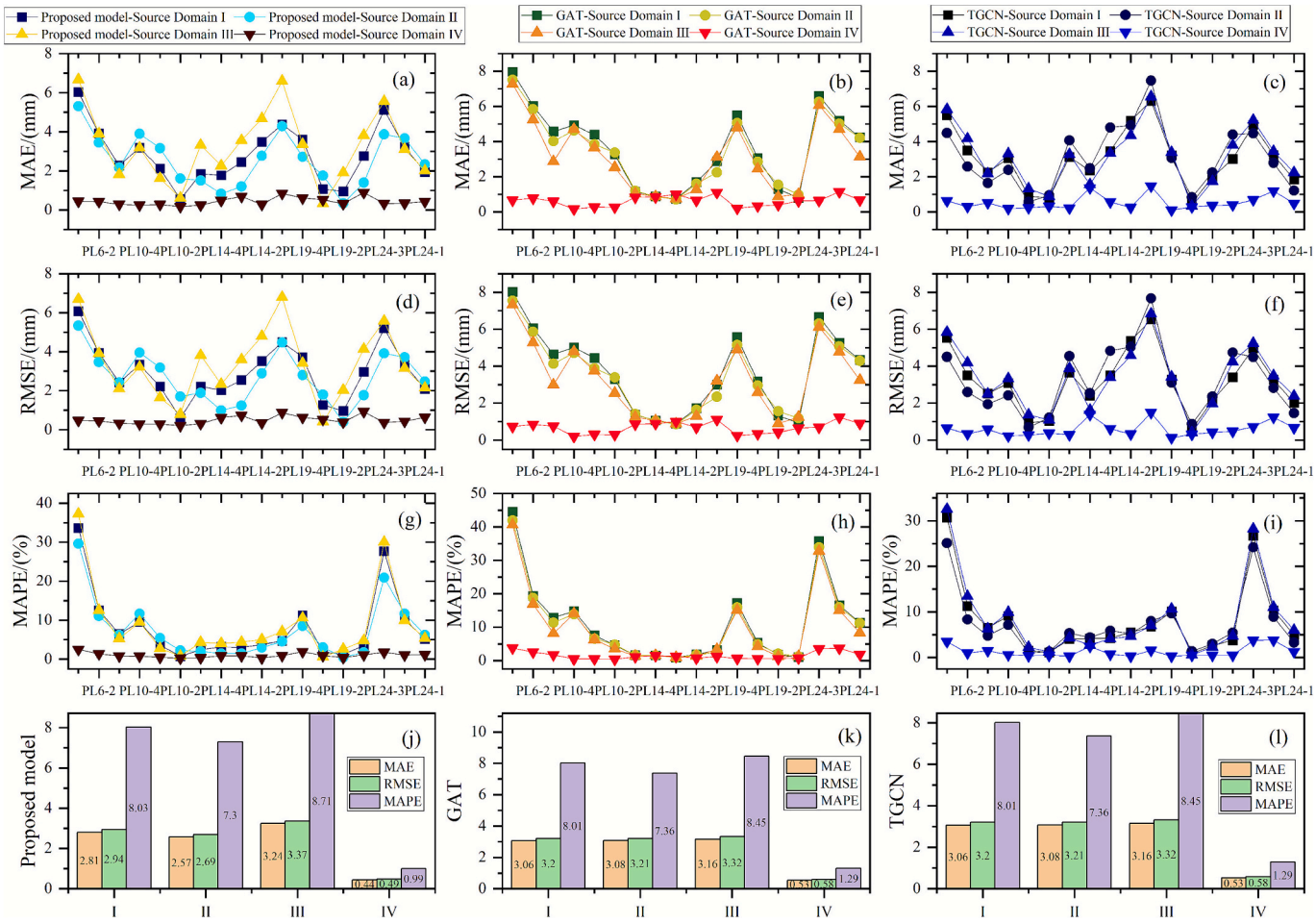

Fig. 9. Comparative analysis of cross-domain transfer of the proposed model, GAT, and TGCN on different source domains.

features, f_S^i, f_T^i refers to the feature from the source domain F_S or target domain F_T, respectively, $\Phi(\bullet)$ denotes the feature mapping function, which maps the input features into the RKHS, and $\|\bullet\|$ represents the norm, typically referring to either the Euclidean norm or the norm in RKHS.

After completing feature alignment, the target domain feature F_T are fed into the Shared Feature Decoder (SFD) $g(\bullet)$ to achieve full-domain digital twin prediction $\widehat{\delta}$ of the dam structure:

$$\widehat{\delta} = g(F_T) \tag{19}$$

While learning domain-invariant feature representations is essential, ensuring prediction accuracy for both the source and target domains is equally critical. Therefore, the proposed method employs a total loss function consisting of three components: prediction loss, physics-informed loss, and cross-domain loss. The minimization of the cross-domain loss L_{domain} aims to encourage alignment of the shared feature distributions across domains, thereby enabling the learning of more robust domain-invariant representations.

$$L_{domain} = l_{COCAL}(F_S, F_T) + l_{MMD}(F_S, F_T) \tag{20}$$

The minimization of the prediction loss improves the model's ability to fit the deformation response in both the source and target domains:

$$L_{prcd} = \lambda_S L_{prcd}^S + \lambda_T L_{prcd}^T \tag{21}$$

where L_{prcd}^S and L_{prcd}^T represent the prediction losses for $D_S = \{(x_S^i, y_S^i)\}_{i=1}^{N_S}$ and $D_T = \{(x_T^i, y_T^i)\}_{i=1}^{N_T}$, respectively. λ_S and λ_T are loss balancing parameters.

To incorporate engineering mechanics constraints into the digital twin, a physics-informed loss L_{phys} is designed, comprising two terms: physical smoothness and spatial gradient consistency. Physical smoothness loss constrains the predicted digital twin field $\widehat{\delta}$ to vary smoothly in space, penalizing large local gradients. Spatial gradient variance loss enforces similarity between the spatial gradient patterns of the target-domain prediction $\widehat{\delta}$ and the source-domain ground truth y^S. The total physics-informed loss is then:

$$L_{phys} = \lambda_{sm} L_{smooth} + \lambda_{gr} L_{grad} \tag{22}$$

$$L_{smooth} = \frac{1}{|\varepsilon|} \sum_{i,j \in \varepsilon} \left\| \widehat{\delta}_i - \widehat{\delta}_j \right\|_2^2 \tag{23}$$

$$L_{prcd} = \nabla_s \widehat{\delta} - \nabla_s y^s \tag{24}$$

where λ_{sm} and λ_{gr} are loss balancing parameters, ε is the set of spatially adjacent finite element nodes (as defined by the adjacency matrix), and ∇_s denotes the discrete spatial gradient operator defined over ε.

By jointly optimizing these three losses, high-accuracy digital twin modelling can be achieved under cross-domain scenarios. The total loss function is defined as follows:

$$L_{tatal} = \lambda_1 L_{prcd} + \lambda_2 L_{phys} + \lambda_3 L_{domain} \tag{25}$$

where λ_1, λ_2, and λ_3 are loss balancing parameters.

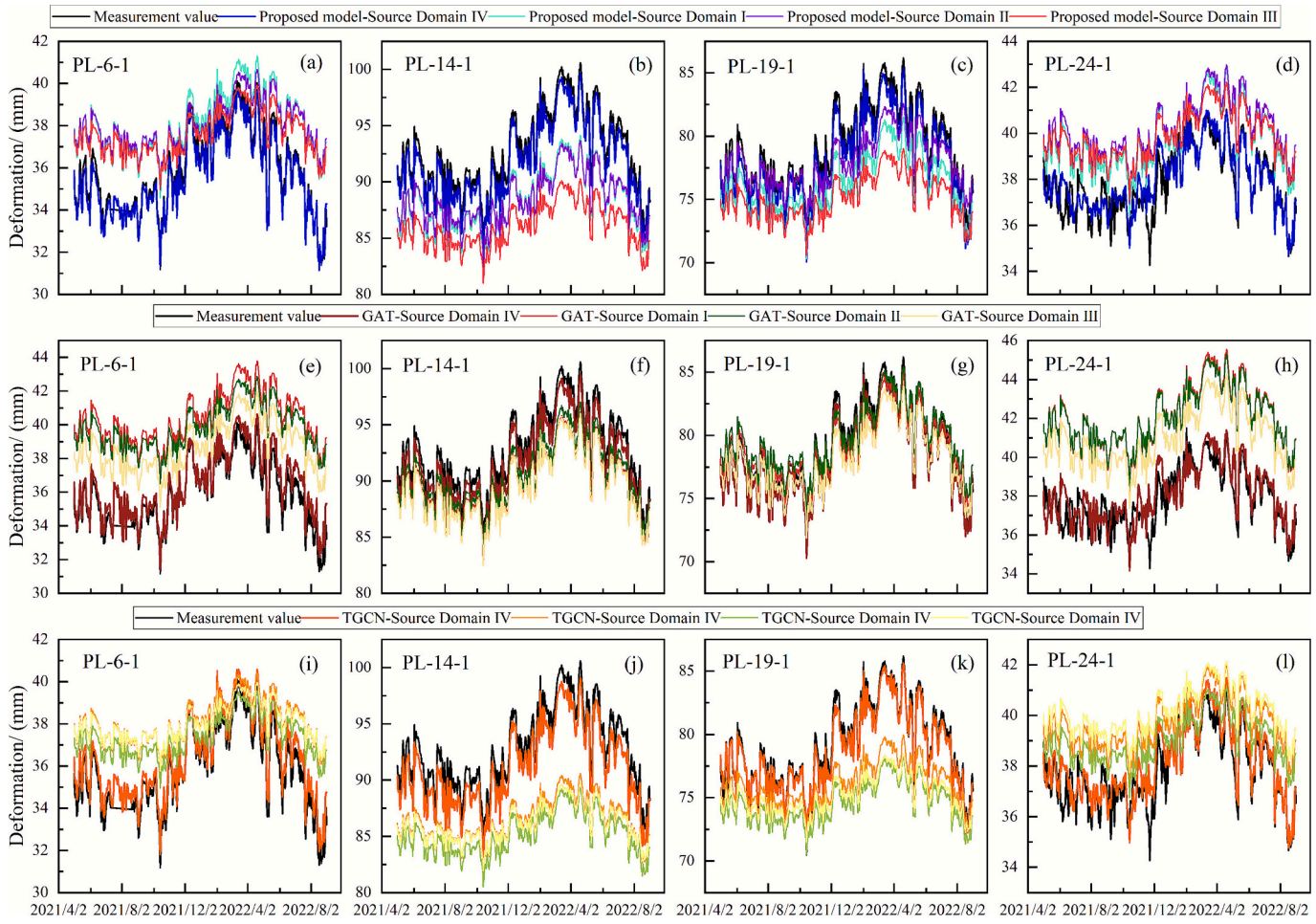

Fig. 10. Comparison of measurement process lines for cross-domain transfer of the proposed model, GAT, and TGCN on different source domains.

3. Case study

3.1. Project overview

In this study, as shown in Fig. 4, the DS Dam and PB Dam, both located on the mainstream of the Dadu River in Sichuan Province and serving as important components of cascade hydropower projects, are selected as validation cases. The DS Dam is a 210 m-high double-curvature arch dam, forming the 14th cascade hydropower station on the Dadu River and functioning as a daily regulation reservoir. The dam has a total installed capacity of 2,600 MW, with a normal reservoir water level of 1,130.00 m and a dead water level of 1,120.00 m. Safety monitoring is conducted using the plumb-line method, with multiple monitoring points arranged at typical dam sections (#6, #10, #14, #19, #24) as well as in the grout galleries at the dam abutments. Monitoring results show that the reservoir exhibits a high frequency but small amplitude of daily water-level fluctuations. The deformation of the dam is influenced by both the periodic effects of temperature and water level and a long-term gradual trend.

The PB Dam is a rockfill dam with a gravel-soil core wall, with a maximum height of 186 m and a total storage capacity of 5.337 billion m^3, serving as an annual regulation reservoir. The dam crest elevation is 856.00 m, with a crest length of 540.5 m. It adopts a composite slope configuration, with a cross-section consisting of a gravel-soil core wall, filter layers, transition zones, and outer rockfill zones. Cable-type horizontal displacement meters are installed within the dam body to monitor layered displacement in the downstream transition zone and the rockfill zone of the core wall. A total of 40 sensors, organized into eight sets (CH1-CH40), are deployed, with some installed in Section 210 + at elevations of 808 m, 758 m, and 731 m. Monitoring data reveal a clear periodic horizontal deformation pattern strongly correlated with reservoir water-level fluctuations, as well as an irreversible gradual deformation trend, reflecting the time-dependent behaviors of the fill materials, such as creep and consolidation.

3.2. Datasets description

As shown in Table 1, this study employs a total of eight datasets. Specifically, the DS Dam includes five source-domain datasets and one target-domain dataset, while the PB Dam includes one source-domain dataset and one target-domain dataset. For the DS Dam, Source Domain I is constructed by inverting only transient parameters. Source Domain II involves inversion of both transient and creep parameters, targeting monitoring points PL10-2, PL10-1, PL14-1, PL14-2, PL14-3, PL14-4, PL19-1, and PL19-2. Source Domain III also inverts both transient and creep parameters, but targets monitoring points PL10-1, PL14-1, PL14-2, PL14-3, PL14-4, and PL19-1. Source Domain IV is generated by combining Source Domains I, II, and III, whereas Source Domain V extends Source Domain IV by incorporating extreme hydraulic conditions (check flood level and dead water level) and extreme temperature conditions (mean temperature ± 20 °C). The target-domain dataset for the DS Dam is constructed from longitudinal displacement monitoring data.

The cross-domain differences between the source and target domains are compared through kernel density estimation and spatial correlation analysis, as shown in Fig. 5. The results indicate that, except for a few

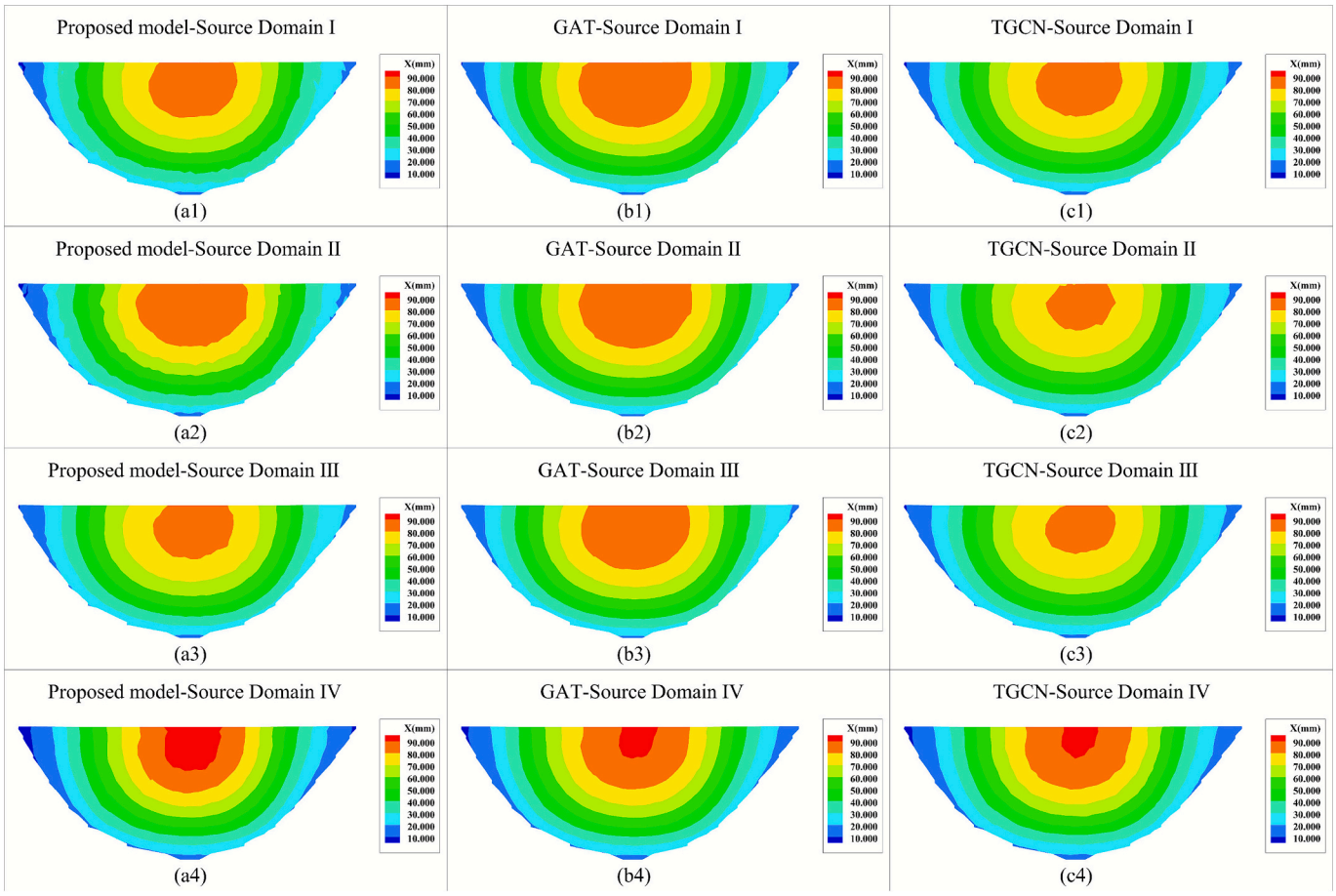

Fig. 11. Comparison of 3D visualization of cross-domain transfer of the proposed model, GAT, and TGCN on different source domains.

monitoring points, the two domains are generally adaptable in terms of distribution, although an overall shift still exists. in terms of spatial correlation, the differences between each source domain and the target domain are relatively significant. The source- and target-domain datasets for the PB Dam are built following the same methodology as for the DS Dam, but with only one parameter inversion scheme. All datasets are divided into training, validation, and test sets in a 0.7:0.1:0.2 ratio. The model inputs are deformation monitoring points or their corresponding finite element mesh nodes, and the outputs are the full-domain finite element mesh nodes.

3.3. Baseline models

To comprehensively validate the effectiveness of the proposed cross-domain adaptive modelling method, this study selected ten representative comparison models, which were categorized into two groups based on task characteristics. The first group consists of four ablation models, designed to evaluate the contribution of each key component: Ablation Model I removes both the CORAL and MMD losses; Ablation Model II removes the GAT-based feature encoder; Ablation Model III removes both the CORAL and MMD losses as well as the GAT-based feature encoder; Ablation Model IV [43,44] is based on Ablation Model III but uses Source Domain I (while all other models use Source Domain IV). The second group consists of six mainstream baseline models for dam structural health monitoring, including LSTM [58], GRU [64], GCN [57], Transformer [49], GAT [57], and TGCN [65], which serve as benchmarks against traditional machine learning and deep learning approaches. Table 2 presents the parameter configurations of all models, where the baseline model parameters are selected under computational constraints to ensure optimal configurations, thereby guaranteeing fairness and consistency in experimental conditions.

4. Results

This experiment is conducted under the Windows 10 system environment using PyCharm IDE and Pytorch2.5.1 based on Python3.10. The server configuration follows: CPU is Intel Core i7-13700, GPU is Nvidia RTX 4070SUPER. In the experiment, the learning rate is set to 0.001, the batch size is 64, the optimizer selects the Adam optimizer to update the network parameters, and the number of iterations is set to 500. To quantitatively analyze the evaluation performance of each model, three evaluation metrics, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are chosen in this paper to evaluate the difference between real and predicted flows [43].

4.1. Ablation experiments of the proposed model

To validate the effectiveness of the proposed GAT-based architecture, domain-adaptive feature alignment method, and source-domain construction approach, four ablation models were designed for comparative analysis, as shown in Fig. 6, Fig. 7, and Fig. 8. Fig. 6 presents the comparison of MAE, RMSE, and MAPE across all monitoring points for the proposed model and the ablation models. The results indicate that the proposed model achieves the highest accuracy at the vast majority of monitoring points, with an average MAE of only 0.44 mm and an average MAPE of 0.99 %, demonstrating excellent prediction precision and confirming the effectiveness of the proposed method in enhancing digital twin accuracy. In terms of the average MAE, the proposed model outperforms Ablation Models I, II, III, and IV by 22.81

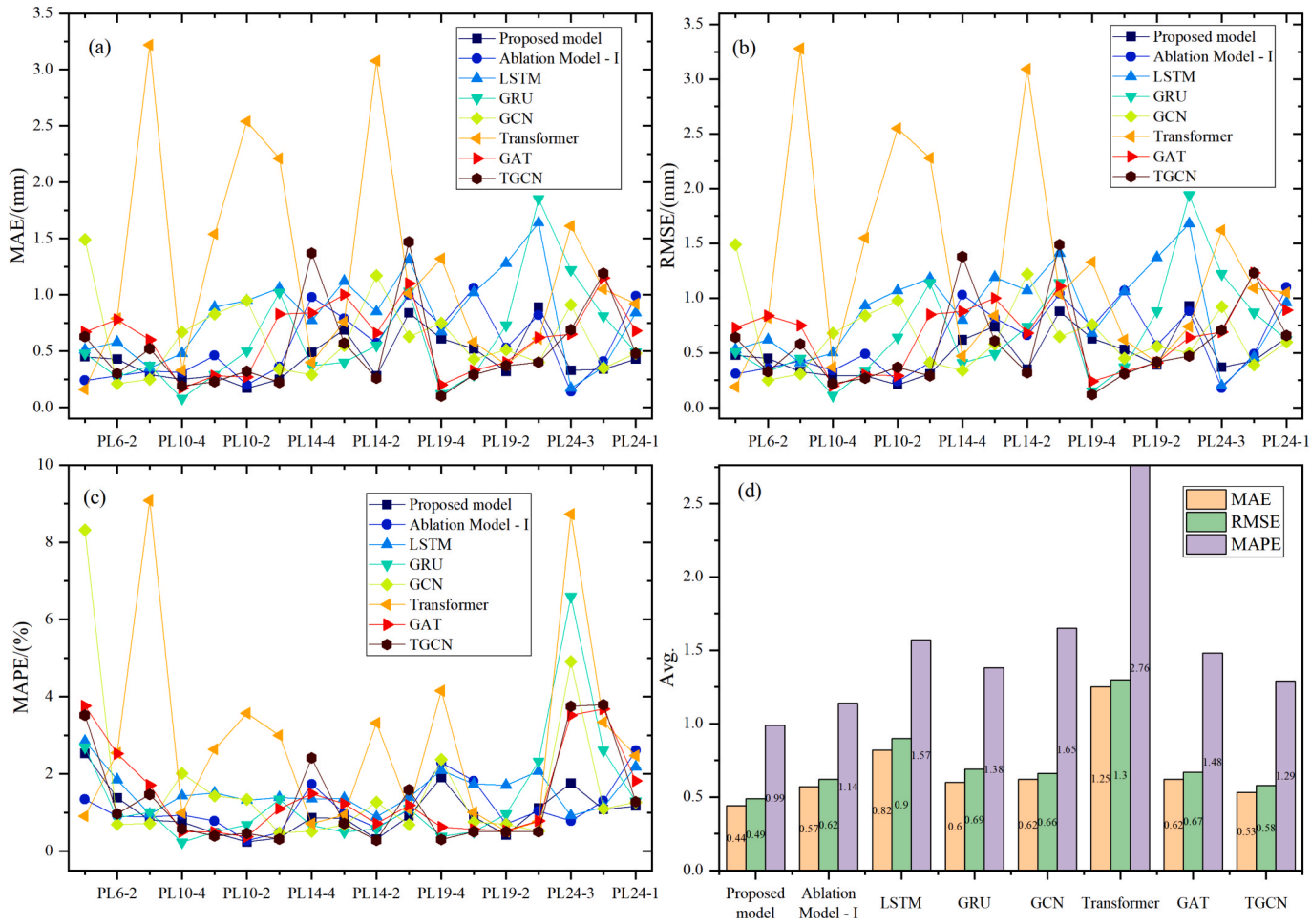

Fig. 12. Comparison of accuracy between the proposed model and the baseline models on DG Dam.

%, 43.59 %, 45.68 %, and 87.02 %, respectively, further validating the contribution of each key component. Comparison with Ablation Model IV shows that source-domain construction contributes more significantly to improving digital twin accuracy than network architecture optimization or feature alignment, highlighting the critical importance of constructing a high-quality source domain in digital twin modelling.

Fig. 7 presents the comparison of predicted time-series values between the proposed model and the ablation models on the test set. The results indicate that the proposed model, as well as Ablation Models I, II, and III, can fit the measured values with high accuracy. However, compared with Ablation Model IV, the proposed model effectively mitigates prediction bias caused by distribution discrepancies between numerical simulations and field measurements. Furthermore, during periods of high water-level fluctuations, the proposed model demonstrates superior fitting performance relative to Ablation Models I, II, and III, indicating enhanced generalization capability and accuracy under extreme conditions, and further validating the effectiveness of each ablation module. Fig. 8 shows the three-dimensional visualization results of the proposed model and the ablation models on May 1, 2021. Through internal slices at multiple and representative angles, it can be observed that the proposed model adheres to physical principles in the digital twin of the high arch dam, while the other models also exhibit satisfactory performance in terms of fitting physical behavior.

4.2. Analysis of the impact of source-domain construction methods

To further investigate the impact of source-domain construction methods on the proposed cross-domain digital twin modelling framework, cross-domain transfer modelling comparisons were conducted for Source Domains I, II, III, and IV under three neural network architectures (the proposed model, GAT, and TGCN), as shown in Fig. 9, Fig. 10, and Fig. 11. Fig. 9 presents the modelling accuracy of the three models across all monitoring points for different source domains. The results indicate that the overall accuracy of the three models is consistent across the source domains. Among single-source domains, Source Domain II achieves the best performance, demonstrating that incorporating both transient deformation and creep parameters, while increasing the number of inversion target monitoring points, enhances consistency between the source and target domains, thereby improving cross-domain transfer modelling accuracy. More importantly, Source Domain IV exhibits the best performance in cross-domain transfer. For the proposed model, the average MAE on Source Domain IV is reduced by 84.34 %, 82.88 %, and 86.42 % compared with Source Domains I, II, and III, respectively, indicating that multi-source domain transfer can significantly improve digital twin modelling accuracy relative to single-source domains.

Fig. 10 shows the comparison of predicted time-series values for the three models across different source domains. The results demonstrate that all three models can capture the variation of measured values well for the four source domains. However, only for Source Domain IV do the predicted values at all monitoring points almost perfectly match the measurements, indicating that multi-source domain transfer effectively mitigates distribution shifts between numerical simulation source domains and monitoring target domains. Fig. 11 presents three-dimensional visualizations of cross-domain modelling of downstream displacement on May 1, 2021, for the three models across different

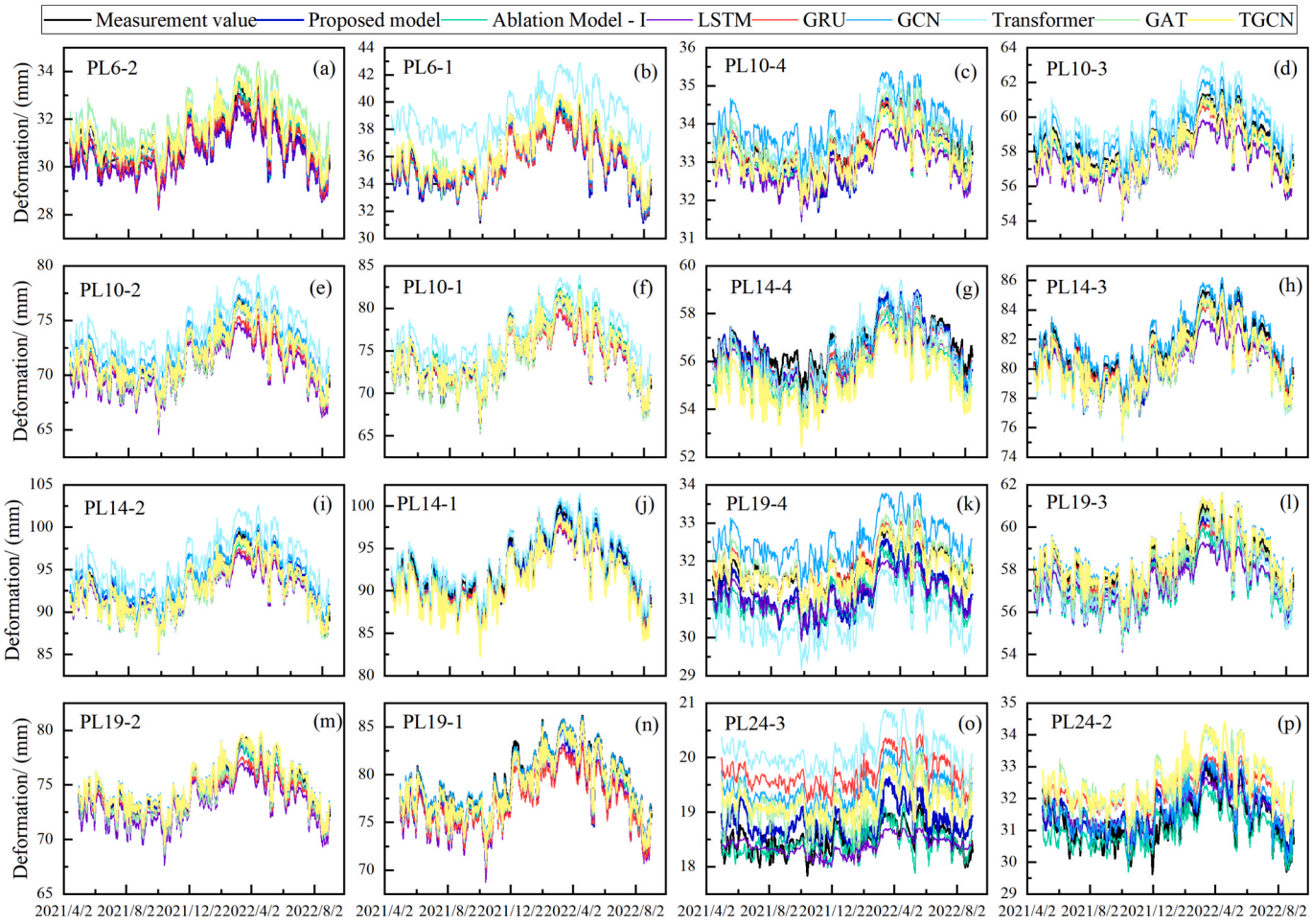

Fig. 13. Comparison of the proposed model and baseline models on the test set process line of DG Dam.

source domains. It can be observed that all three models satisfy physical principles across the four source domains, with differences only in the magnitude of predicted values, further validating the effectiveness of multi-source domain transfer for high-accuracy digital twin modelling.

4.3. Comparison between the proposed model and baseline models

To further validate the effectiveness of the proposed model, six baseline models, including LSTM, GRU, GCN, Transformer, GAT, and TGCN, were selected for cross-domain transfer modelling on Source Domain IV of the DG Dam and the source domain of the PB Dam. The results are presented in Figs. 12–17. Fig. 12 and Fig. 15 show the comparison of modelling accuracy at each monitoring point between the proposed model and the baseline models for the DG Dam and PB Dam, respectively. The results indicate that the proposed model achieves the highest modelling accuracy for both the concrete and embankment dams, with the average MAE reduced by 29.03 % and 14.05 % compared with GAT, respectively, demonstrating strong cross-domain generalization capability. Additionally, Ablation Model I also performs relatively well on certain metrics, suggesting that the proposed network architecture itself contributes positively to modelling effectiveness. The results further reveal that GAT and TGCN achieve relatively good accuracy on the DG Dam but perform moderately on the PB Dam, whereas LSTM performs well on the PB Dam but moderately on the DG Dam. These findings indicate that the cross-domain generalization ability of different neural network architectures can be uncertain, while a high-quality source domain can significantly enhance the stability and accuracy of cross-domain modelling. As presented in Table 3, the statistical robustness of the findings is further verified by repeating the training five times with different random seeds for the proposed model and the best-performing baselines (Ablation Model I, LSTM, and GAT) on the DG Dam. The proposed model yielded mean ± 95 % confidence intervals of MAE = 0.35 ± 0.09 mm and RMSE = 0.40 ± 0.09 mm on the DG Dam dataset, outperforming all baselines. This indicates that the performance gains are statistically significant and the model exhibits high robustness against random fluctuations in network initialization and optimization. Such repeated experimental validation confirms that the proposed cross-domain adaptive framework maintains consistent accuracy across stochastic trials, reinforcing the model's reliability for real-world digital twin deployment in dam monitoring and safety assessment.

Fig. 13 and Fig. 16 present the comparison of predicted time-series values between the proposed model and the baseline models on the test sets of the DG Dam and PB Dam, respectively. The results indicate that for the DG Dam, even with multi-source domain construction, baseline models such as GCN and Transformer still exhibit prediction bias, highlighting the significant impact of neural network architecture and feature alignment on cross-domain modelling accuracy and generalization. The three-dimensional visualization in Fig. 14 further shows that, except for the Transformer, the other models adhere well to physical principles, suggesting that neural network architecture not only affects temporal prediction but may also result in insufficient learning in spatial distributions. For the PB Dam, the comparison of predicted time-series values between the proposed model and the baseline models indicates that, due to the use of a single-source domain and the complex cross-domain characteristics of the embankment dam structure, the

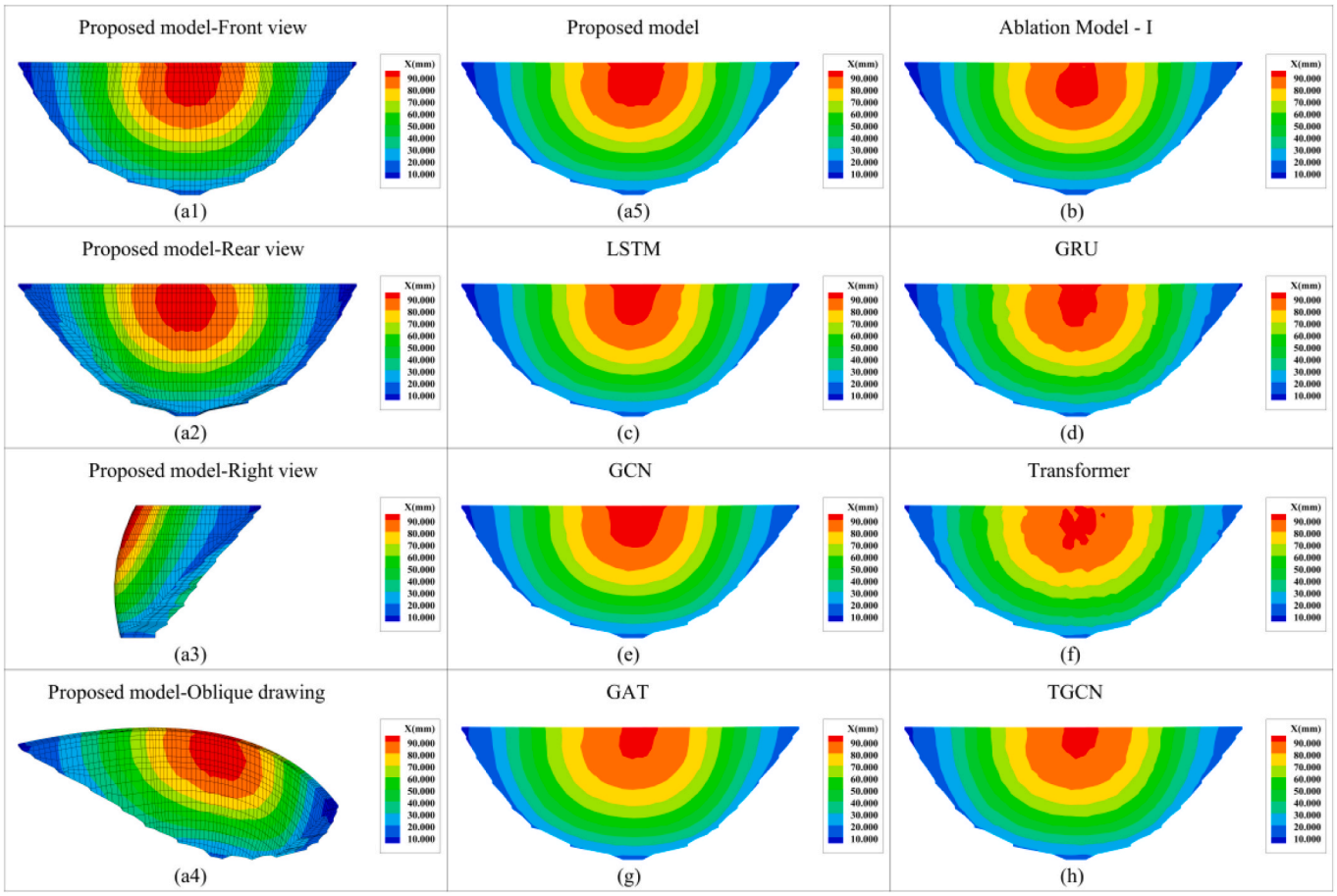

Fig. 14. Comparison of 3D visualization between the proposed model and the baseline model on the DG Dam.

predictions deviate more noticeably from measurements. This demonstrates that improving neural network architecture alone is insufficient to achieve high-accuracy cross-domain digital twins, and that multi-source domain construction and feature alignment remain key factors for enhancing accuracy. Combined with Fig. 17, it can be observed that all models generally satisfy physical principles in two-dimensional cross-section predictions. However, GAT exhibits poorer smoothness, indicating that the generalization capability of the models still requires improvement and that the critical modules affecting generalization performance within the neural network need to be clarified.

4.4. The proposed model has generalization ability under extreme working conditions

To further evaluate the generalization capability of the proposed model under extreme conditions, two categories of extreme scenarios were designed in this study. The first category concerns water level conditions, including the check flood level and the dead water level. The second category relates to temperature conditions, in which the average temperature was increased or decreased by 20 ℃. Based on these settings, a source domain V that includes extreme conditions was constructed and compared with source domain IV, which does not incorporate extreme conditions. The validation results are shown in Figs. 18–20. As illustrated in Fig. 18, the model was evaluated under both routine monitoring and extreme conditions for source domain IV and source domain V. The results indicate that after introducing samples from extreme conditions, the model achieved a significant improvement in prediction accuracy under routine conditions, with the average MAE, RMSE, and MAPE reduced by 20.45 %, 13.95 %, and 22.22 %, respectively. Meanwhile, under source domain IV, the prediction accuracy under extreme conditions decreased at several individual monitoring points such as PL16-1 and PL19-1, suggesting limitations in the model's generalization ability. In contrast, under source domain V, the overall accuracy of the model was further improved, with the average MAE, RMSE, and MAPE reduced by 33.33 %, 23.26 %, and 30.16 %, respectively, and the improvements under extreme conditions were more pronounced. As shown in Fig. 19, the predicted process lines under both source domain IV and source domain V were closely aligned with the measured values and the simulated values under extreme conditions, with only minor discrepancies. Furthermore, Fig. 20 presents the three-dimensional visualization results, which demonstrate that the model predictions were consistent with physical laws and that the overall differences remained within a controllable range. In summary, the proposed model exhibits strong generalization performance under extreme conditions, and the inclusion of extreme condition samples in the source domain can effectively enhance both prediction accuracy and cross-domain adaptability.

5. Discussion

Building upon previous LSTM-FEM coupled digital twin models [43,44] for high arch and earth-rock dams, this study redefines the boundary between data-driven prediction and physics-based simulation by introducing a cross-domain adaptive learning framework. The primary limitation of existing dam digital twins lies in the significant discrepancy between numerical simulations and field monitoring data, which constrains prediction accuracy and generalization under extreme or rare operating conditions. To address this, multiple high-fidelity source domains are constructed by systematically varying material properties, boundary conditions, and load scenarios in the finite element

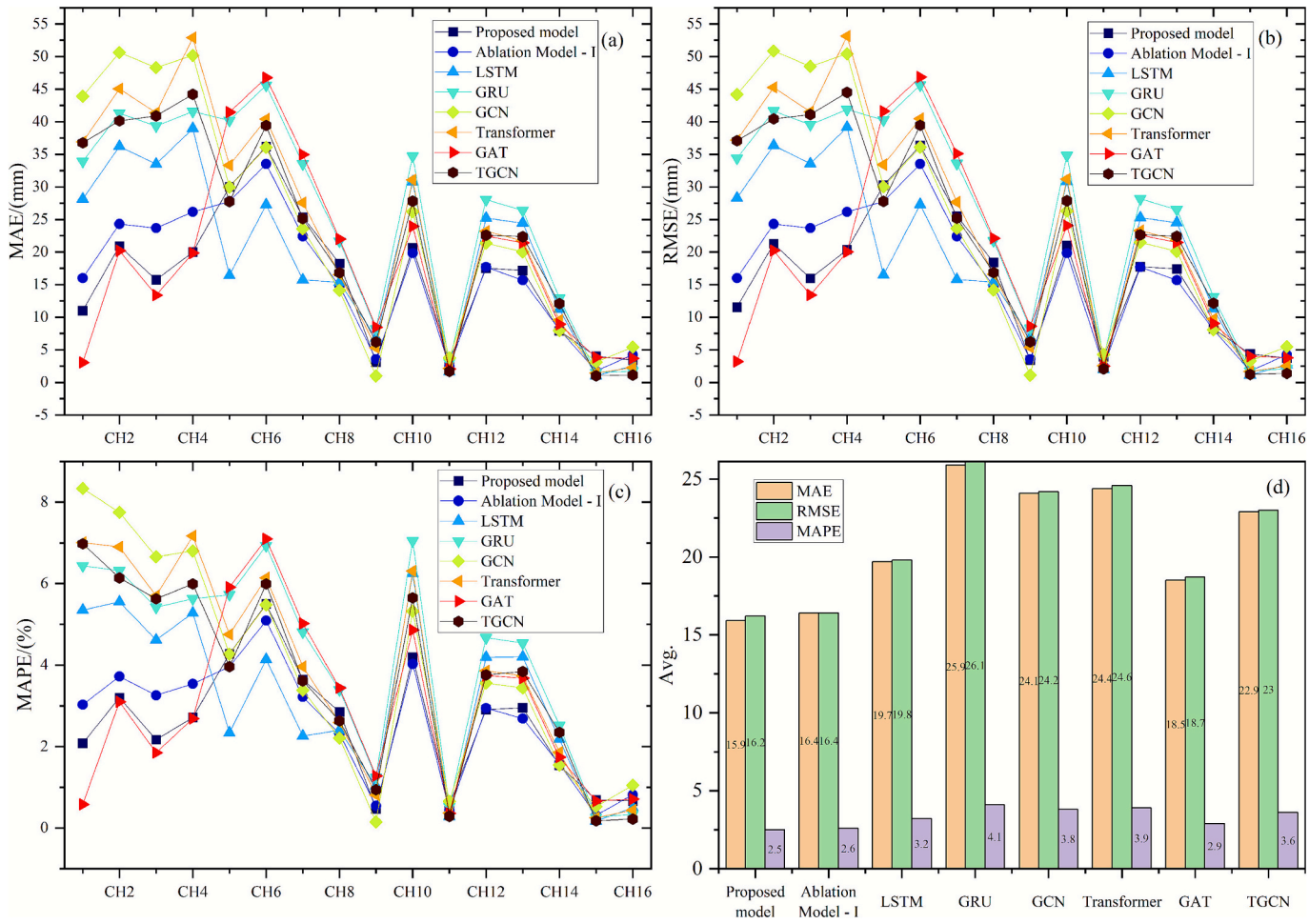

Fig. 15. Comparison of accuracy between the proposed model and the baseline models on PB Dam.

simulations. This approach effectively expands the feature space of the source domain, reducing the domain gap and enhancing the model's ability to generalize across different structural and operational conditions.

In addition to improving predictive accuracy, the proposed graph attention-gated network architecture introduces a new level of interpretability. By capturing spatial dependencies among monitoring points and structural nodes, the attention mechanism quantifies the relative influence of each node on the predicted deformation or stress field, while the gating mechanism regulates the propagation of information over space. This enables the model to maintain stability and physical consistency even under sparse sensing or noisy measurement conditions. Importantly, this interpretable design allows engineers to gain insights into the internal decision-making process of the digital twin, bridging the gap between "black-box" neural network predictions and physical understanding.

From a broader perspective, this study establishes a generalizable framework for physics-informed domain adaptation in dam digital twins, integrating multi-source learning with mechanistic reasoning. The findings demonstrate that high-quality, diversified source domains and adaptive feature alignment are critical for achieving both accuracy and robustness, particularly under extreme loading or failure conditions. Moreover, the framework provides a foundation for future extensions that incorporate emerging technologies. For instance, generative AI methods, such as diffusion-based models, could augment source domains with synthetic but physically consistent deformation fields, improving robustness in data-scarce or hazardous scenarios. Alignment with Industry 5.0 principles, including human-centric AI and industrial metaverse applications, could transform the digital twin into an interactive, human-in-the-loop decision-support tool, enabling real-time feedback, immersive visualization, and collaborative diagnostics. Finally, the framework may be extended to semi-physical commissioning environments, allowing real-time, closed-loop validation by integrating live sensor data with simulated responses. These developments collectively position the proposed digital twin framework as a scalable, interpretable, and robust approach for next-generation cyber-physical-human systems in hydraulic engineering.

6. Conclusion

This study proposes a novel cross-domain adaptive digital twin modelling framework for high arch and earth-rock dams, integrating GAT, gating mechanisms, and feature alignment strategies to improve prediction accuracy and generalization across domains. The main findings are summarized as follows:

(1) The proposed model achieves superior prediction accuracy compared with ablation and baseline models. On average, the model reduces MAE by 22.81 %, 43.59 %, 45.68 %, and 87.02 % relative to ablation variants, and by 29.03 % and 14.05 % compared with state-of-the-art baselines on concrete and earth-rock dams, respectively. These results confirm the effectiveness of the GAT-based architecture, source-domain construction, and cross-domain feature alignment in enhancing digital twin accuracy.

(2) The study demonstrates that constructing high-quality source domains is crucial for cross-domain modelling. Multi-source domain transfer (Source Domain IV) outperforms single-source domains,

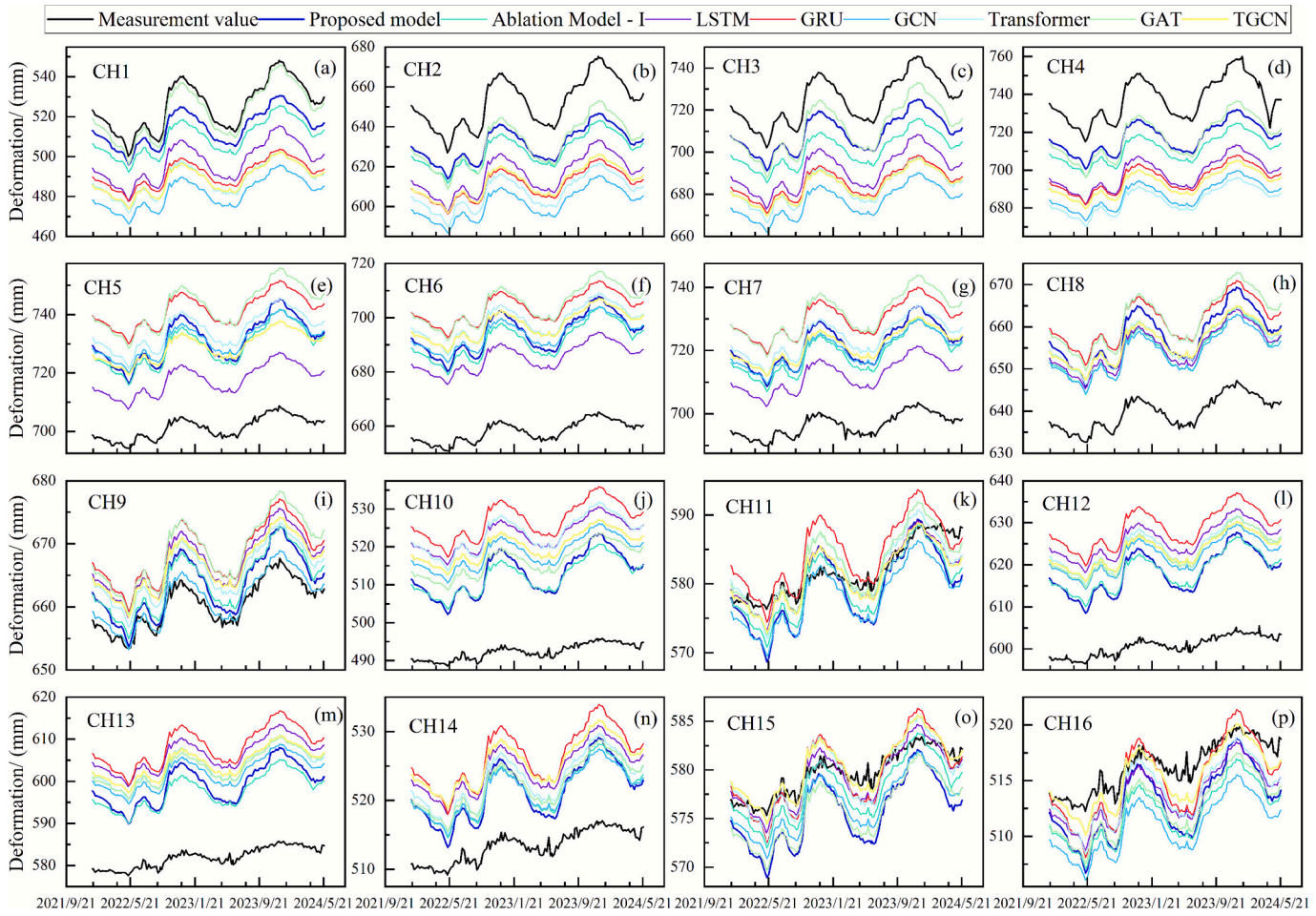

Fig. 16. Comparison of the proposed model and baseline models on the test set process line of PB Dam.

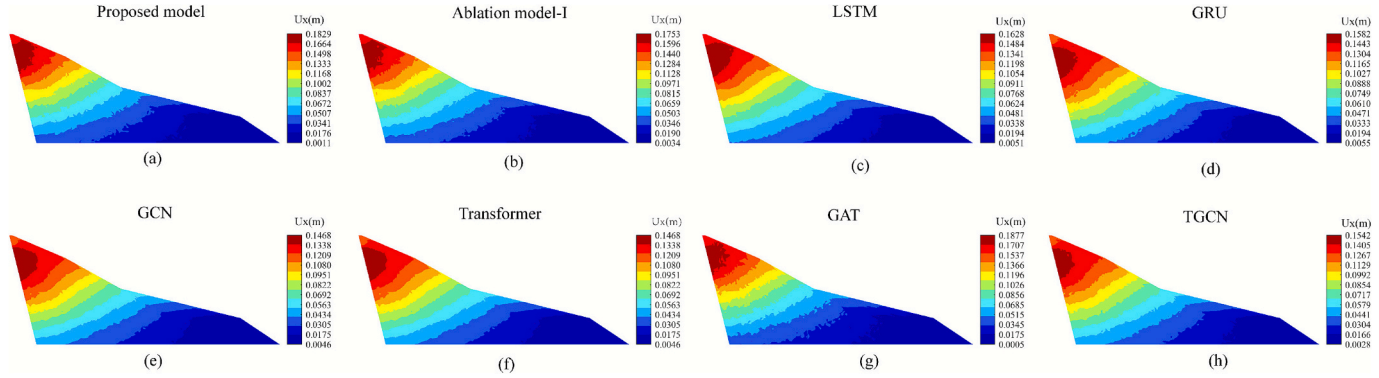

Fig. 17. Comparison of 2D visualization between the proposed model and the baseline model on the PB Dam.

reducing prediction errors by over 80 % and significantly mitigating distribution discrepancies between simulation and monitoring data. This highlights the importance of incorporating diverse simulation scenarios in digital twin modelling.

(3) Validation under extreme water level and temperature conditions confirms that the proposed model possesses strong generalization capability. Incorporating extreme-condition samples (Source Domain V) further improves accuracy, reducing MAE, RMSE, and MAPE by 33.33 %, 23.26 %, and 30.16 %, respectively, and effectively alleviating prediction degradation at critical monitoring points.

Overall, the proposed framework provides an effective and physically consistent digital twin modelling approach for dam structures, offering robust adaptability across domains and under extreme operating conditions. Future research will focus on enhancing the fusion of multi-source monitoring information, quantifying cross-domain differences between numerical simulations and actual monitoring, verifying the generalization of structural damage, integrating domain knowledge, and constructing intelligent agents to support large-scale, real-time dam safety monitoring.

CRediT authorship contribution statement

Jichen Tian: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation. **Ruijie Yu:** Writing – review &

Table 3
Statistical summary (mean ± 95 % CI) of evaluation metrics across random seeds.

Model	Proposed model			Ablation Model − I			LSTM			GAT		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PL6-3	0.24 ± 0.18	0.27 ± 0.18	1.34 ± 1.03	0.63 ± 0.59	0.69 ± 0.59	3.55 ± 3.28	0.45 ± 0.07	0.49 ± 0.04	2.49 ± 0.41	0.93 ± 0.44	0.98 ± 0.42	5.2 ± 2.44
PL6-2	0.28 ± 0.13	0.31 ± 0.12	0.9 ± 0.43	0.64 ± 0.49	0.7 ± 0.48	2.06 ± 1.55	0.57 ± 0.07	0.62 ± 0.06	1.81 ± 0.24	0.73 ± 0.43	0.8 ± 0.4	2.36 ± 1.39
PL6-1	0.28 ± 0.05	0.33 ± 0.05	0.78 ± 0.15	0.81 ± 0.82	0.88 ± 0.77	2.26 ± 2.29	0.57 ± 0.21	0.65 ± 0.22	1.56 ± 0.58	0.86 ± 0.24	0.97 ± 0.22	2.41 ± 0.68
PL10-4	0.3 ± 0.08	0.34 ± 0.07	0.9 ± 0.23	0.3 ± 0.07	0.35 ± 0.1	0.9 ± 0.21	0.58 ± 0.15	0.61 ± 0.14	1.73 ± 0.46	0.47 ± 0.54	0.51 ± 0.53	1.42 ± 1.6
PL10-3	0.31 ± 0.08	0.33 ± 0.07	0.53 ± 0.13	0.34 ± 0.18	0.38 ± 0.19	0.59 ± 0.31	0.84 ± 0.22	0.9 ± 0.17	1.43 ± 0.37	0.52 ± 0.21	0.53 ± 0.2	0.9 ± 0.35
PL10-2	0.17 ± 0.01	0.22 ± 0.03	0.23 ± 0.02	0.4 ± 0.28	0.45 ± 0.27	0.57 ± 0.4	0.86 ± 0.16	1 ± 0.11	1.19 ± 0.22	0.55 ± 0.25	0.56 ± 0.24	0.76 ± 0.35
PL10-1	0.41 ± 0.25	0.46 ± 0.24	0.54 ± 0.34	0.35 ± 0.03	0.42 ± 0.02	0.46 ± 0.05	1.19 ± 0.18	1.31 ± 0.17	1.56 ± 0.24	0.8 ± 0.43	0.82 ± 0.42	1.06 ± 0.57
PL14-4	0.28 ± 0.19	0.34 ± 0.25	0.49 ± 0.34	0.95 ± 0.39	0.99 ± 0.38	1.67 ± 0.68	1 ± 0.31	1.02 ± 0.3	1.76 ± 0.54	0.83 ± 0.6	0.86 ± 0.59	1.46 ± 1.06
PL14-3	0.35 ± 0.29	0.38 ± 0.31	0.43 ± 0.36	0.71 ± 0.16	0.75 ± 0.17	0.87 ± 0.2	1.16 ± 0.27	1.21 ± 0.25	1.43 ± 0.34	0.86 ± 0.48	0.87 ± 0.47	1.06 ± 0.58
PL14-2	0.27 ± 0.03	0.32 ± 0.04	0.3 ± 0.03	0.44 ± 0.11	0.53 ± 0.12	0.47 ± 0.12	0.9 ± 0.2	1.08 ± 0.14	0.95 ± 0.22	0.83 ± 0.26	0.85 ± 0.26	0.89 ± 0.28
PL14-1	0.67 ± 0.24	0.7 ± 0.24	0.72 ± 0.26	0.84 ± 0.31	0.9 ± 0.31	0.9 ± 0.33	1.48 ± 0.26	1.55 ± 0.22	1.59 ± 0.28	0.88 ± 0.51	0.9 ± 0.49	0.95 ± 0.55
PL19-4	0.32 ± 0.25	0.36 ± 0.23	1 ± 0.79	0.38 ± 0.32	0.42 ± 0.31	1.18 ± 1.02	0.73 ± 0.12	0.75 ± 0.11	2.29 ± 0.38	0.42 ± 0.39	0.46 ± 0.38	1.31 ± 1.22
PL19-3	0.36 ± 0.14	0.37 ± 0.14	0.61 ± 0.25	0.63 ± 0.38	0.67 ± 0.37	1.09 ± 0.66	0.92 ± 0.19	0.96 ± 0.18	1.57 ± 0.33	0.34 ± 0.06	0.34 ± 0.06	0.58 ± 0.11
PL19-2	0.41 ± 0.24	0.47 ± 0.22	0.55 ± 0.33	0.56 ± 0.23	0.65 ± 0.26	0.74 ± 0.3	1.04 ± 0.25	1.12 ± 0.24	1.38 ± 0.34	0.61 ± 0.2	0.62 ± 0.2	0.82 ± 0.27
PL19-1	0.7 ± 0.35	0.75 ± 0.34	0.88 ± 0.44	1.19 ± 0.7	1.28 ± 0.71	1.49 ± 0.88	1.57 ± 0.32	1.63 ± 0.3	1.98 ± 0.4	0.63 ± 0.25	0.64 ± 0.25	0.79 ± 0.32
PL24-3	0.41 ± 0.15	0.44 ± 0.14	2.2 ± 0.79	0.54 ± 0.42	0.58 ± 0.43	2.93 ± 2.29	0.18 ± 0.05	0.22 ± 0.06	0.99 ± 0.26	1.07 ± 0.64	1.1 ± 0.63	5.79 ± 3.47
PL24-2	0.34 ± 0.08	0.41 ± 0.08	1.1 ± 0.25	0.53 ± 0.26	0.61 ± 0.26	1.68 ± 0.81	0.33 ± 0.05	0.43 ± 0.05	1.05 ± 0.15	1.15 ± 0.25	1.23 ± 0.24	3.68 ± 0.79
PL24-1	0.28 ± 0.14	0.39 ± 0.22	0.75 ± 0.37	0.84 ± 0.14	1 ± 0.1	2.21 ± 0.35	0.92 ± 0.18	1.04 ± 0.17	2.41 ± 0.47	0.88 ± 0.19	1.1 ± 0.19	2.35 ± 0.51
Avg.	0.35 ± 0.09	0.4 ± 0.09	0.79 ± 0.17	0.62 ± 0.15	0.68 ± 0.15	1.42 ± 0.56	0.85 ± 0.15	0.92 ± 0.13	1.62 ± 0.26	0.74 ± 0.1	0.79 ± 0.1	1.88 ± 0.51

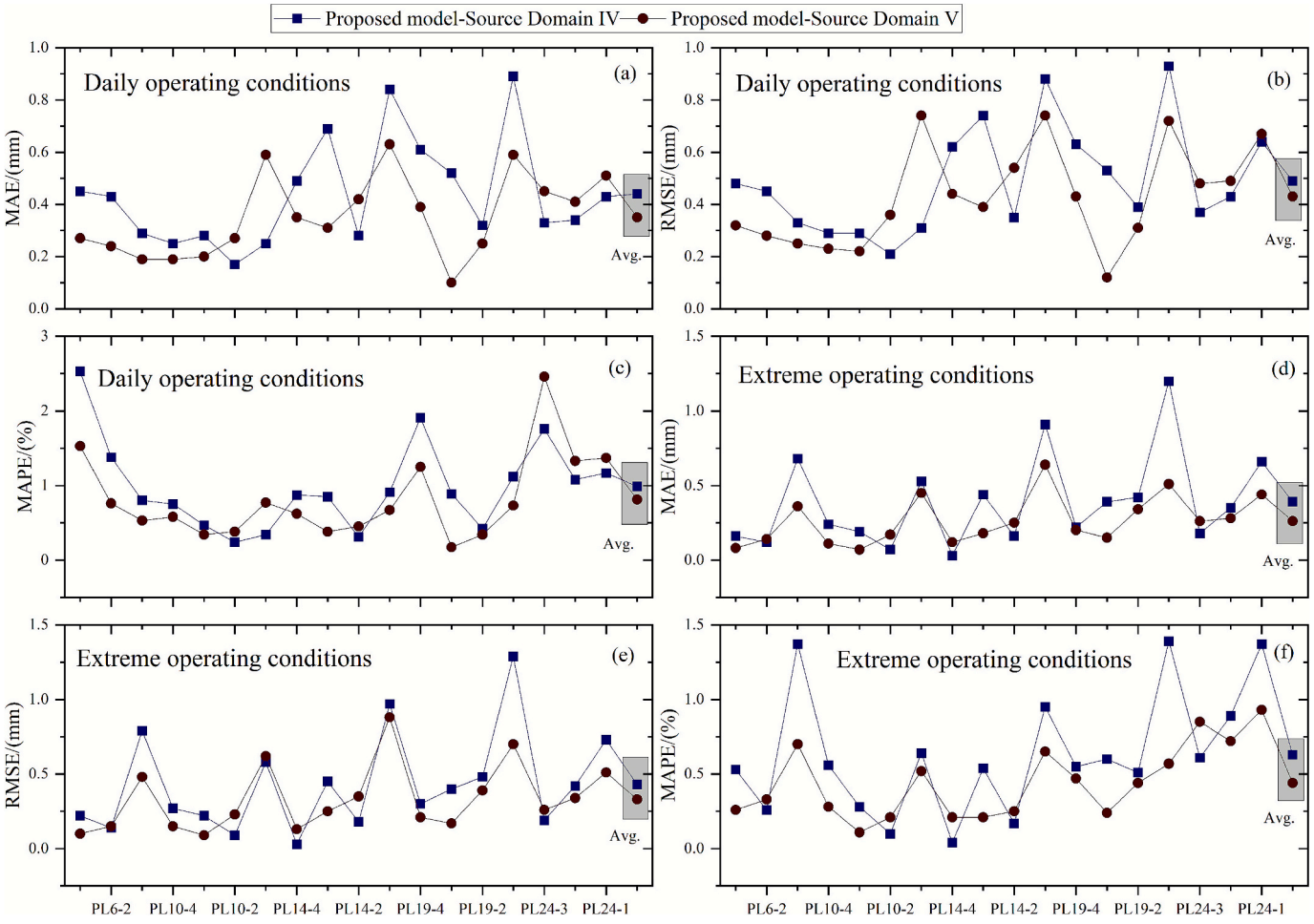

Fig. 18. The generalization of the proposed model in different source domains under extreme operating conditions.

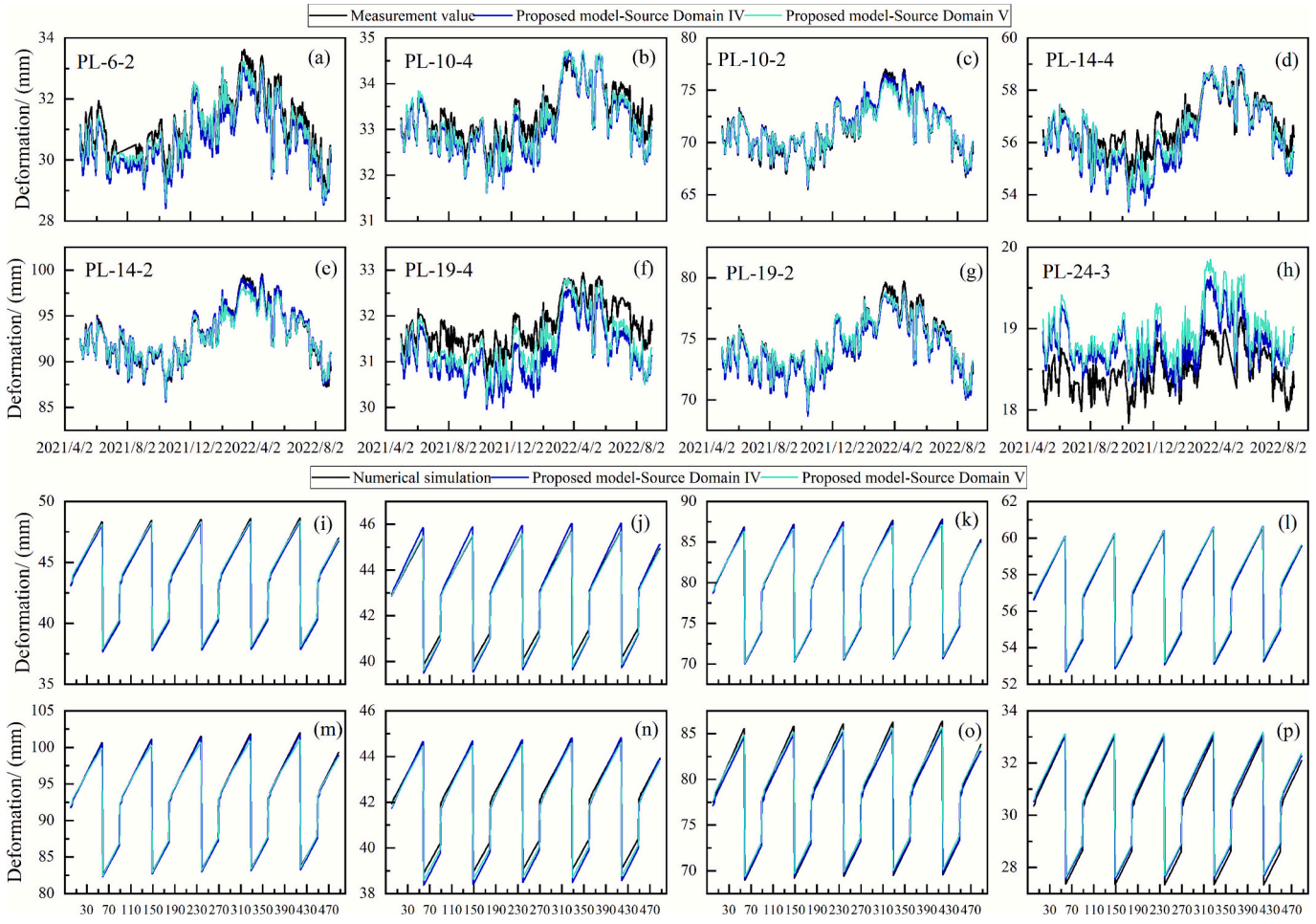

Fig. 19. Comparison of the measurement process of the proposed model under extreme and daily operating conditions. Daily operating conditions: (a) PL6-2 (b) PL-10–4 (c) PL10-2 (d) PL14-4 (e) PL14-2 (f) PL19-4 (g) PL19-2 (h) PL24-3 Extreme operating conditions: (i) PL6-2 (j) PL-10–4 (k) PL10-2 (l) PL14-4 (m) PL14-2 (n) PL19-4 (o) PL19-2 (p) PL24-3.

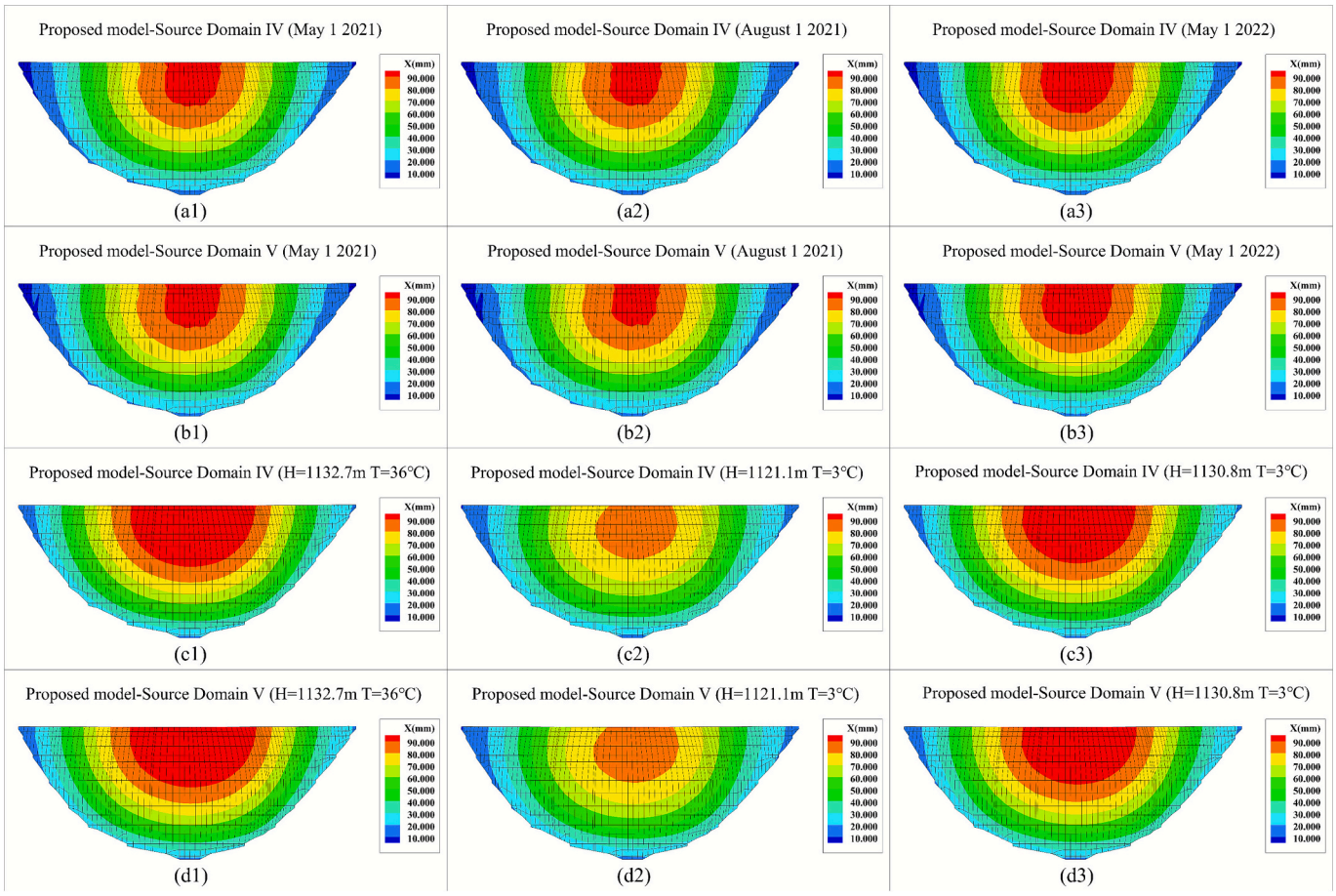

Fig. 20. Comparison of 3D visualization of the proposed model under extreme and daily operating conditions.

editing, Software, Methodology, Data curation. **Chen Chen:** Writing – review & editing, Resources, Funding acquisition. **Yanling Li:** Writing – review & editing, Supervision, Investigation, Funding acquisition. **Huibao Huang:** Writing – review & editing, Software, Resources, Data curation. **Jiankang Chen:** Writing – review & editing, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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